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Monitoring Technologies- Continuous Glucose Monitoring, Mobile Technology, Biomarkers of Glycemic Control

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ABSTRACT

It is recognized that traditional measures of glucose control (such as hemoglobin A1c [A1c]) provide little information regarding the need for day-to-day changes in therapies. While intermittent self-monitored blood glucose (SMBG) provides additional information with which to make treatment decisions, significant barriers to its use exist, such as inconvenience and lack of timely and regular feedback. Furthermore, important information regarding glucose trends may be missed. Continuous glucose monitoring (CGM) has well-established reliability and efficacy in terms of improving A1c, reducing hypoglycemia, and improving the time in target glucose range. Automated insulin delivery (AID) systems, which link CGM with algorithm-driven insulin delivery, are now widely available and represent the preferred insulin delivery method in type 1 diabetes. Due to the massive continual stream of data presented to patients and providers, it is paramount that the data are organized in a standardized way and that communication of data is streamlined using patients' mobile devices where available and within the existing clinic infrastructure. Systems that provide immediate feedback to patients and decision support tools for patients and providers have demonstrated superior outcomes compared to routine SMBG alone. Alternate markers of glucose control may provide complementary information about glucose levels and are particularly useful when the A1c is not reliable. This chapter will review the latest evidence for use of glucose monitoring technologies, and biomarkers of glycemic control. For complete coverage of all related areas of Endocrinology, please visit our on-line FREE web-text, WWW.ENDOTEXT.ORG.

INTRODUCTION

The current technology for monitoring of glucose levels has been well established since the 1980s. This practice is beneficial to patients with diabetes from both a clinical and an economic standpoint when used optimally. Knowledge of the glucose levels that are measured can allow a patient to select an appropriate dose of insulin or implement dietary or other lifestyle changes to regulate their glucose levels. Expert groups provide recommendations for glucose targets, including hemoglobin A1c (A1c), self-monitored blood glucose (SMBG), and interstitial glucose (continuous glucose monitoring [CGM]) (1,2). Although targets vary, expert groups recommend individualization based upon risk of hypoglycemia, comorbidities, life expectancy, and individual characteristics that may affect long-term benefit. The American Diabetes Association (ADA) recommends assessing overall glucose levels using the A1c and/or CGM metrics such as % Time in Range (TIR, the % of time spent 70-180 mg/dL), or the Glucose Management Indicator (GMI), which is an estimate of A1c that is derived from a 14-day CGM report for routine assessment of glucose levels.

The landscape of glucose monitoring technologies is expanding and rapidly changing. For a full review of glucose monitoring technologies, the reader is referred to one of many excellent reviews referenced throughout this chapter. Several trends are emerging in glucose monitoring and will be reviewed in more detail in this chapter:

> **CGM**: This practice is becoming more widely established as evidence supporting its use has accumulated. The data available through CGM can permit significantly more fine-tuned adjustments in insulin dosing and other therapies than spot testing from SMBG can provide. CGM technologies for automatic collection of data have spurred interest in noninvasive glucose monitoring as an additional tool for obtaining information about glucose levels.
>
> **Closed loop control (CLC):** Also known as an “artificial” or “bionic” pancreas, this technology links CGM with automated insulin delivery (AID). The first steps toward CLC are now in use.
>
> **Mobile Technology and Decision Support**: In recent years, increasing connectivity between glucose monitoring technologies and mobile devices has facilitated ongoing improvements in self-care and communication of data.
>
> **Alternate Markers of Glucose Control:** Finally, the use of additional analytes besides glucose is still being established.

This chapter analyzes the technology, benefits, and challenges with the use of intermittent SMBG and CGM, mobile technology and decision support, and alternate biomarkers of glycemic control.

CONTINUOUS GLUCOSE MONITORS

CGM measures glucose levels (typically interstitial glucose) continuously and updates the glucose level display periodically (usually every 1-5 minutes). Most CGMs consist of 1) a monitor to display the information (in many cases, this is the patient’s mobile device), 2) a sensor that is usually inserted into the subcutaneous tissue, and 3) a transmitter that transmits the sensor data to the monitor. The transmitter may be self-contained with the sensor in a single unit or separate. Previously, all devices were approved for adjunctive use only, requiring fingerstick glucose monitoring to guide therapy and perform calibrations. All major personal CGM systems are now approved for stand-alone (nonadjunctive) use, meaning that under specified conditions they may be used to make treatment decisions without a confirmatory blood glucose measurement. Newer technologies have also eliminated the requirements for calibration of CGM with a fingerstick glucose. The accuracy of all commercially available CGMs is still the lowest in the hypoglycemic range, which is where the need for sensitivity and specificity is great in terms of serving as an alarm for hypoglycemia.

CGM can provide both retrospective as well as real-time information to detect: 1) hypoglycemic and hyperglycemic excursions; 2) impending hypoglycemia; and 3) wide fluctuations in glucose levels, also known as glycemic variability. 24-hour manufacturer support is available for all FDA approved CGM devices. Use of CGM can help both the patient and their medical provider make fine-tuned adjustments to medications and provide insight to the patient on behavioral changes to achieve glycemic control. Additionally, current efforts to link CGM measurement with AID systems have progressed incrementally toward a fully functional artificial pancreas. CGM systems can be divided according to their intended use as professional CGM (which is a clinic-owned device and provides either retrospective or real-time glucose data) and personal CGM (which is patient-owned and provides real-time glucose data).

PROFESSIONAL CGM

Professional CGM describes CGM data that are obtained via healthcare provider owned equipment. It does not necessarily provide the glucose results in real time, but downloads the readings after they have been collected, similar to a 24-hour cardiac Holter monitor that provides information about cardiac rhythms after they have occurred. This allows the health care provider to obtain relatively unbiased glucose patterns during typical everyday life. The

Endocrine Society recommendations state that professional CGM may be of benefit in adults with diabetes to detect nocturnal hypoglycemia, dawn phenomenon, postprandial hyperglycemia and to assist in management of diabetes therapies (3). Professional CGM is more readily covered by payers than personal CGM, particularly among individuals who are not using insulin. However, interpretation of both personal and professional CGM reports by qualified healthcare professionals may be billed and reimbursed on a monthly basis.

As the use of and coverage of personal devices has expanded, the use of professional CGM has declined. The FreeStyle Libre Pro, Medtronic iPro2, and the Dexcom G6 Pro were discontinued in 2026. Where blinded data are needed, personal devices could be used with notifications silenced or not paired to an app to achieve a similar result. For the Dexcom, a G7 device can be configured for professional use by setting up blinded mode on a clinic owned receiver. Personal systems will be discussed in more detail later (see "Personal [Real-time] Continuous Glucose Monitoring"). In a meta-analysis of 22 articles, professional CGM resulted in a greater reduction in A1c (-0.28%, 95% CI -0.36% to -0.21%, $P < 0.00001$) as well as TIR (5.59%, 95% CI 0.12 to 11.06, $P = 0.05$) compared to usual practice (4).

Analysis of Retrospective Data

Data from all CGM devices can be studied retrospectively after downloading (5). It is recommended that diet, activity, symptoms, and insulin data are collected during professional CGM to assist with interpretation, either via patient diary, direct entry of events into the device, or use of an accompanying app, depending on the system. Three time periods should be analyzed. These are:

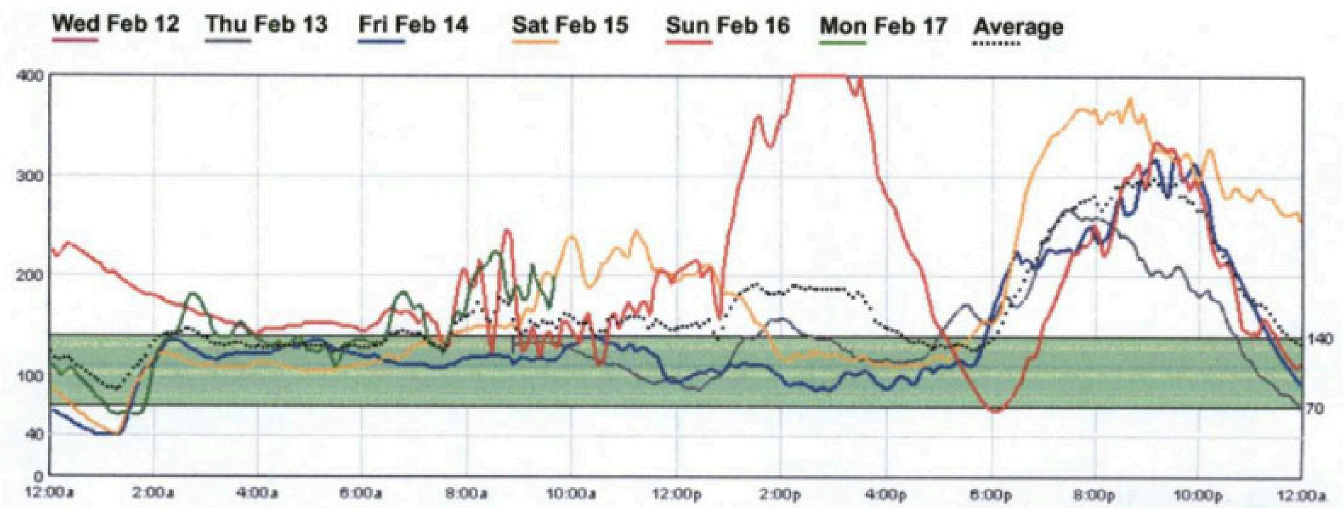
> **Overnight**: Out-of-target overnight glucose levels can be modified by adjusting the basal insulin dose.
>
> **Pre-prandial Period**: Out-of-target pre-prandial glucose levels can be modified by adjusting the previous meal bolus, meal, or exercise pattern.
>
> **Post-prandial Period**: Out-of-target postprandial glucose levels can be modified by adjusting the immediate meal bolus, meal, or exercise pattern.

In certain special situations, targets may need to be adjusted. Other important elements of a professional CGM analysis are shown in Table 1. An example of a patient who used CGM is presented in Figure 1. The CGM demonstrated high glucose levels from 6:00 PM to 11:00 PM post-supper and low glucose levels from 12:00 AM to 2:00 AM. Recognition of these patterns allowed appropriately timed treatment interventions.

Table 1.

Elements of Professional Continuous Glucose Monitoring Analysis

Overall Control
Mean Glucose
Glucose Variability (Standard Deviation, Coefficient of Variation)
Daily Detail
Diurnal Patterns: dawn phenomenon, overnight
Meal effects
Correction
Exercise effects
Other patterns (workdays vs. weekend, menstrual cycles)
Hypoglycemia
Precipitating factors
Corresponding meter glucose (recognition)



	Wed Feb 12	Thu Feb 13	Fri Feb 14	Sat Feb 15	Sun Feb 16	Mon Feb 17	Average / Total
# Sensor Values	0	182	288	288	288	116	1,162
Highest	N/A	267	322	379	400	224	400
Lowest	N/A	69	40	41	66	61	40
Average	N/A	150	138	179	202	140	166
Standard Dev.	N/A	50	62	88	81	39	75
MAD %	N/A	14.5	24.1	17.6	18.0	10.3	17.5
Correlation	N/A	N/A	N/A	0.97	0.94	N/A	0.95
# Valid Calibrations	0	3	2	4	4	1	14
Designation	S		X			X	

X: Use Clinical Judgment S: No Sensor Data C: No Calibration BG's

Figure 1.

CGM tracing of a patient whose glucose levels were high from 6:00 PM to 11:00 PM post-supper and low from 12:00 AM to 2:00 AM.

Ambulatory Glucose Profile

Prior to 2017, each CGM manufacturer structured their downloads differently, creating inconsistency in clinical interpretation. The Ambulatory Glucose Profile (AGP) was standardized for CGM use in 2017 when an international expert consensus panel recommended a common set of CGM metrics and the AGP format as the standard reporting structure, driven by the International Diabetes Center and endorsed by major diabetes organizations (6).

The AGP (Figure 2) is a standardized reporting format for glucose data that was developed by an expert panel of diabetes specialists and is customized for insulin pumps or injection therapy (7). The universal report is intended to simplify and facilitate interpretation of otherwise complex and lengthy reports with varying terminology. By 2019, the standardized AGP has been actively implemented in clinical practice. A single page report that the medical team can view and file into a patient's electronic medical record and that can be used as a shared decision-making tool for people with diabetes was considered to be of great value (8). The 2023 international consensus on CGM metrics for clinical trials introduced updates to the AGP layout (9). A stacked bar graph visually summarizes glucose metrics with discrete percentages for different glucose categories. Consistent color coding (green for target, red for extreme values) improves clarity and safety interpretation. The 2026 ADA Standards of Care reaffirmed this structure, endorsing a three-panel AGP format that displays the following (10):

- CGM metrics including percentage of values in the target range, above and below targets, as well as an assessment of glucose variability.
- 24-hour glucose profile obtained from the past 14 days, displaying median glucose and variability with color-coded zones (yellow for high, red for low, green for target range).

Daily glucose profiles over 14 days to identify differences based on variable routines (e.g., weekends vs. weekdays).

This AGP format provides a standardized framework for CGM data review, enabling clinicians to identify glycemic patterns and inform therapy adjustments.

Core CGM Metrics

The eight ADA-recommended core CGM metrics, building on international consensus, provide a standardized framework for assessing glycemic control (10):

Mean glucose: Average glucose over the monitoring period.

GMI: Converts mean glucose into an estimated A1c using the equation: GMI (%) = 3.31 + 0.02392 × mean glucose (mg/dL). GMI appears in the AGP and aids clinicians in interpreting CGM data in relation to traditional A1c measurements. It is accepted by Healthcare Effectiveness Data and Information Set as a quality metric (11).

Glucose coefficient of variation (%CV): Reflects glucose variability; Target: ≤36% (values <33% may offer added protection in those on insulin or sulfonylureas).

Time in range (TIR): Percent of time in 70-180 mg/dL. Target: >70% for most individuals.

Time below range (<70 mg/dL): Reflects level 1 hypoglycemia. Goal: <4% for most individuals.

Time <54 mg/dL: Reflects level 2 hypoglycemia. Goal: <1%.

Time above range (>180 mg/dL): Reflects hyperglycemia. Goal: <25% for most individuals.

Time >250 mg/dL: Reflects severe hyperglycemia. Goal: <5%.

Additional recommendations apply to specific populations:

Older/high-risk adults: TIR >50%, time <70 mg/dL <1%, time >250 mg/dL <10%.

Pregnancy (type 1 diabetes, T1D): TIR >70% (63-140 mg/dL), time <63 mg/dL <4%, time <54 mg/dL <1%.

The ADA and the European Association for the Study of Diabetes (EASD) emphasize that even a 5% increase in TIR is clinically meaningful (10,12). TIR strongly correlates with A1c and has been linked to risk of microvascular complications (13). GMI, while not always equivalent to lab A1c due to individual biological variation, offers useful insight when interpreted alongside A1c.

Composite Metrics

As a measure of the quality of glycemia, the TIR, similar to the A1c is limited in its assessment of hypoglycemia. Multiple composite metrics have thus been reported(14). However, the use of multiple metrics increases complexity and is subject to issues with collinearity. The Glycemia Risk Index (GRI) is a composite metric that was developed using input from 330 clinical experts who analyzed 14-day tracings from 225 adults with diabetes (15). GRI more heavily weights very high or very low glucose values and correlates with clinician rankings more closely than TIR or time below range (%time < 70 mg/dL, TBR) alone.

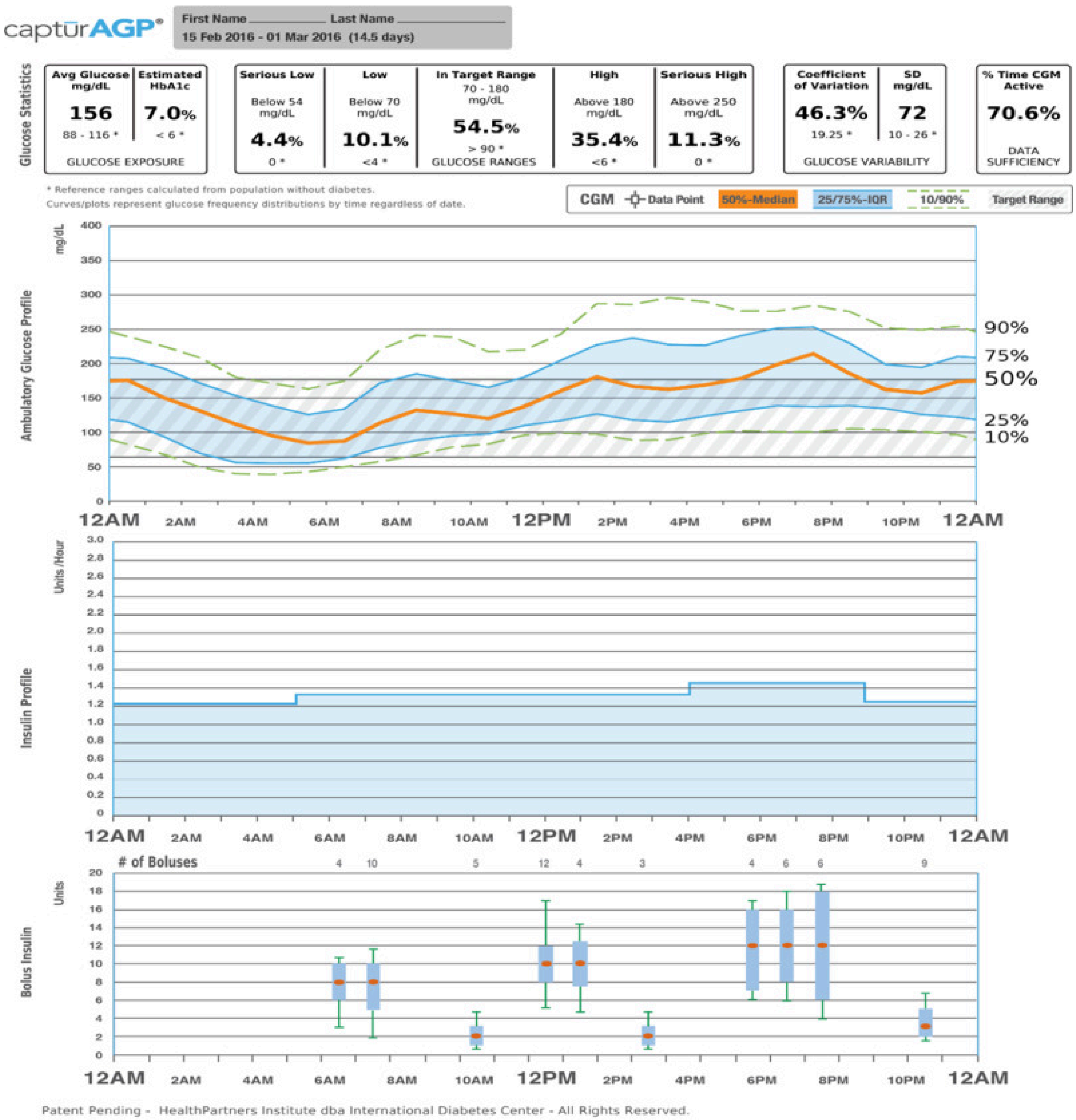


Figure 2.

Ambulatory Glucose Profile for Insulin Pumps.

Glucose Statistics: Metrics include mean glucose, estimated A1c, glucose ranges, coefficient of variation and standard deviation.
Glucose Profile: Daily glucose profiles are combined to make a one-day (24-hour) picture. Ideally, lines would stay within grey shaded area (target range).
Orange: median (middle) glucose line.
Blue: area between blue lines shows 50% of the glucose values.
Green: 10% of values are above (90% top line) and 10% are below (10% bottom line).

Insulin Profile Graph: Shows basal insulin pump settings over a 24-hour period.

Bolus Insulin Graph: Combines all bolus insulin doses into one graph to make a one day (24-hour) picture. Each box on the graph covers 60 minutes of doses.
Orange: median (middle) dot.
Blue: shaded box shows 50% of the bolus dosages in the hour.
Green: lines above and below the shaded box (whiskers) show how many of the bolus dosages per hour were between 75 - 90% and between 10 - 25%.

PERSONAL REAL-TIME CGM (rtCGM) OR INTERMITTENTLY SCANNED CGM (isCGM)

RtCGM devices not only display the current glucose every few minutes but may also alert the patient to impending (projected alert) or actual (threshold alert) hyperglycemia or hypoglycemia or rate of change in glucose. By comparison, isCGM requires patient interaction with the device to obtain readings but may still provide alerts for hypoglycemia or hyperglycemia. While few head-to-head studies are available, some studies suggest greater reduction in hypoglycemia and improvement in TIR with rtCGM compared to isCGM in persons with T1D (16,17), even up to 24 months (18).

Over time, accuracy with rtCGM and isCGM has improved substantially (19–21). In fact, most devices, including the Dexcom and Freestyle Libre are approved for stand-alone use, meaning that under specified conditions, the device may be used to make treatment decisions without a confirmatory blood glucose measure. However, the user will still experience a tradeoff between a high alarm sensitivity and specificity for detecting hypoglycemic events, particularly where glucose levels are changing rapidly (Figure 3). Current and recent glucose levels, trend information, and a visual alarm are all presented so that a patient can predict future low or high glucose excursions. Using this information will allow the patient to take actions to spend more time in the euglycemic range and less time in the hypoglycemic or hyperglycemic ranges. This potential decrease in glycemic variability will not necessarily be reflected in an improved A1c which reflects mean glycemic levels.

	High Sensitivity (Minimal false negative rate)	High Specificity (Minimal false positive rate)
Advantages	Safety will be maximized	Sleep and other activities will rarely be unnecessarily disturbed
Disadvantages	False alarms will be common	Some hypoglycemic events will be missed

Figure 3.

Tradeoffs between emphasis on high sensitivity compared to emphasis on high specificity in a hypoglycemic alarm that is part of a continuous glucose monitor.

Evidence- Type 1 Diabetes

Studies may be divided according to background therapies (insulin pump or injection therapy).

STUDIES UTILIZING EITHER INSULIN PUMP OR INJECTIONS AS BACKGROUND THERAPIES

The seven-country GuardControl Study was the first randomized controlled trial (RCT) to ever demonstrate a statistically significant improvement in A1c levels with the use of rtCGM (22). The Guardian RT was used either continuously or biweekly for three months and both regimens were compared to control treatment which did not include use of CGM. At one month and at three months, the continuous users had significantly lower A1c levels than the controls. The biweekly users had intermediate improvement which did not reach statistical significance compared to the outcomes in the control group.

In 2008, the Juvenile Diabetes Research Foundation (JDRF) CGM Study Group evaluated 322 adults and children with T1D (either injection or insulin pump therapy) and A1c 7-10% who were randomized to either rtCGM or usual care (23). RtCGM was associated with a 0.53% reduction in A1c compared to usual care ($p<0.001$) but was only significant among subjects over age 24 due to lack of consistent use in younger patients. Hypoglycemia was infrequent and was not different between groups.

In 2011, 120 children and adults with T1D on insulin pump or injection therapy and A1c <7.5% were randomly assigned to rtCGM (Freestyle Navigator- not available in the US) or masked CGM every other week (24). The time spent in hypoglycemia was reduced over 50% at 26 weeks, and patients spent more time in 70-180 mg/dL range.

In the IMPACT trial, 241 adults with T1D with an A1c ≤7.5% were randomly assigned to Freestyle flash isCGM vs. SMBG. In this group, 68% of the patients were treated with multiple daily injections (MDI) and 32% with Continuous Subcutaneous Insulin Infusion (CSII). The amount of time spent in hypoglycemia was decreased by nearly 90 minutes per day ($p<0.0001$) when patients had access to CGM data (25). It must be noted that this technology does not provide real-time alerts for impending hypoglycemia or hyperglycemia, and data are accessed via a hand-held device on demand. In a small study of patients with hypoglycemia unawareness or recent severe hypoglycemia, rtCGM more effectively reduced the time spent in hypoglycemia compared to isCGM (26).

The CITY study was a randomized study among 153 adolescents and young adults with T1D. CGM resulted in a -0.37% greater reduction in A1c compared to usual care ($p=0.01$) (27). In this study, only 68% of participants used CGM at least 5 days per week in month 6, which is significantly lower than studies reported in adults (28). However, this still represents more than double the adherence reported in the pivotal JDRF study of 2008, which found only ~30% consistent CGM use in the 15-24 age group (23). Moreover, this study utilized an earlier generation CGM which required twice daily calibration.

Among 203 older adults (median age 68) with T1D randomized to CGM or usual care, CGM resulted in less hypoglycemia at 26 weeks (estimated treatment difference 27 minutes/day, $p<0.001$) as well as modest improvement in A1c (estimated treatment difference -0.3%, $p<0.001$) (29). The improvement in hypoglycemia was sustained over 52 weeks, at which point CGM use was still >90% (30).

The France Adoption RCT evaluated the Eversense implantable CGM in adults with T1D across two cohorts: Cohort 1 (A1c >8%) and Cohort 2 (prone to hypoglycemia). In Cohort 2, Eversense reduced time <54 mg/dL by 1.6 percentage points at days 90-120 and 2.6 points by days 150-180 ($p=0.011$), showing sustained benefit. Cohort 1 saw no significant A1c or TIR changes (31).

In a randomized trial of older adults with T1D (mean age 71), participants with ≥2 episodes of blood glucose <70 mg/dL were assigned to usual care or an intervention combining CGM with geriatric principles, which included adjusting glycemic targets based on comorbidities and functional status, and simplifying insulin regimens to reduce treatment burden and hypoglycemia risk (32). The intervention group had a 2.6% median reduction in time <70 mg/dL compared to 0.3% in the control group (p<0.001), with benefits seen in both CGM-naive and experienced users. A1c remained stable demonstrating that individualized, CGM-guided care can reduce hypoglycemia without compromising overall glycemic control.

In the STAR3 study, 485 adults and children with A1c 7.4-9.5% were randomized to sensor-augmented pump therapy (Medtronic Paradigm Revel) or MDI (33). Sensor-augmented pump therapy resulted in better A1c reduction with a between-group difference of 0.6% (p<0.001). Hypoglycemia did not differ between groups, but only short-term CGM data was available for comparison and patients with a history of severe hypoglycemia were excluded.

STUDIES UTILIZING INJECTION THERAPY AS BACKGROUND

In 2016, a 6-month RCT, the DIAMOND study, compared rtCGM (using Dexcom G4 system) versus SMBG in 158 patients with T1D on multi-dose injection therapy and demonstrated a significantly lower A1c (between-group difference 0.6%, p<0.0001), decrease in hypoglycemia (43 minutes vs. 80 minutes per day, p=0.0002) and less glucose variability with rtCGM compared to SMBG. This study did not address hypoglycemia frequency in the two groups (28).

The GOLD trial studied 161 patients with T1D receiving MDI with either rtCGM (Dexcom G4) or standard care in a random order crossover trial. The mean difference in A1c was 0.43% (p<0.001), favoring rtCGM. One subject in the CGM group compared to 5 subjects in the standard care group experienced a severe hypoglycemic event. The percentage of time spent in hypoglycemia numerically favored the CGM group, but statistical analyses were not presented. There was a significant reduction in standard deviation and measures of glucose variability. Overall well-being, diabetes treatment satisfaction, and fear of hypoglycemia improved (34).

In the FLASH-UK study, 156 participants with T1D were randomized to isCGM or usual care (35). The intervention group had a significantly greater reduction in A1c (adjusted treatment difference -0.5%, p<0.001), higher % TIR, and lower % TBR.

An RCT among 104 adults with T1D found that isCGM improved A1c (estimated treatment difference 0.3%, p=0.04) and TIR but not TBR compared to blood glucose monitoring (36).

PATIENTS WITH SEVERE HYPOGLYCEMIA OR HYPOGLYCEMIC UNAWARENESS

Many older studies specifically excluded patients with a history of severe hypoglycemia or were underpowered to detect significant hypoglycemia. Recent studies have examined the use of rtCGM in patients with hypoglycemia unawareness, which is a risk factor for severe hypoglycemia (events requiring outside assistance to treat).

In the HypoCOMPaSS trial, 96 patients with a history of hypoglycemia unawareness determined by the GOLD Score of at least 4 or more were randomly assigned in a 2x2 factorial design to insulin pump or injection therapy, both with access to a bolus insulin calculator, and either rtCGM (Medtronic CGM System) or SMBG. All patients had diabetes education with a goal toward hypoglycemia avoidance (37). The results demonstrated a similar reduction in severe hypoglycemia and improvement in hypoglycemia unawareness and fear of hypoglycemia without a significant treatment interaction between insulin or glucose monitoring interventions. Treatment satisfaction was higher with insulin pump compared to injection therapy but similar between rtCGM and SMBG. In a subsequent analysis, individuals who continued to experience severe hypoglycemia over 24 months tended to have higher baseline rates of asymptomatic hypoglycemia, lower concern for hypoglycemia at baseline and greater prioritization on avoiding hyperglycemia. This study emphasized the need to address cognitive and behavioral barriers in parallel with technology-based interventions (38).

The IN CONTROL trial evaluated patients with T1D and hypoglycemia unawareness receiving either injection or insulin pump therapy in a crossover study comparing rtCGM (Medtronic Paradigm Veo system or SMBG) (39). Hypoglycemia was significantly reduced with rtCGM compared to SMBG (including a 9.8% reduction in events <70 mg/dL and 44% reduction in events <40 mg/dL). Severe hypoglycemic events were significantly reduced but hypoglycemia unawareness was unchanged. In this study, contact with patients was less frequent and sensor use was greater (89% vs. 57%), and there were no insulin adjustment protocols compared to HypoCOMPaSS.

In a smaller study of 52 adults with T1D and problematic hypoglycemia, immediate randomization to CGM was more effective for preventing severe hypoglycemia (39% fewer events, $p<0.05$) than a dedicated hypoglycemia avoidance education program alone (40). CGM also led to greater reduction in A1c (treatment difference -0.47%, $p<0.05$), but impaired awareness was restored in 31% of both groups, supporting the concept that CGM assists in earlier recognition and treatment of impending hypoglycemia as opposed to effecting fundamental change in counterregulatory responses.

A 2024 prospective cohort study followed 60 adults with T1D using the FreeStyle Libre 2 for two years. The study reported a 47% reduction in impaired awareness by Clarke score (24.6% to 11.6%, $p=0.034$), along with significant improvements in time <54 mg/dL, fewer low glucose events, and decreased fear of hypoglycemia, highlighting its value in this patient population (41).

A 2025 post hoc analysis of the WISDM trial examined CGM's impact on impaired hypoglycemia awareness in older adults. While the full Clarke score did not change in the overall population, the SHEF (Severe Hypoglycemia Experienced Factors) subscale improved significantly at 26 and 52 weeks, and the full Clarke score improved among participants with baseline impaired hypoglycemia awareness. These results highlight the importance of the choice of outcome measures when evaluating CGMs effects on awareness (42).

In an RCT of 35 adults with T1D recently treated for severe hypoglycemia, initiation of rtCGM immediately after the event significantly reduced recurrent hypoglycemia over 12 weeks. Compared to SMBG, rtCGM users had lower time spent <55 mg/dL ($p=0.03$), fewer nocturnal hypoglycemic episodes ($p=0.02$), and no recurrent severe hypoglycemic events compared to 4 events in the SMBG group ($p=0.04$), highlighting the clinical value of early initiation of rtCGM after hypoglycemic events (43).

Differences between studies may be explained by differences in populations and the technologies utilized. Therefore, more studies are needed to understand the potential role of background therapy, other technologies, and clinical support.

META-ANALYSES

An older Cochrane review and a separate meta-analysis from 2012 found modest A1c reductions with CGM, with the largest benefits seen among patients using sensor-augmented insulin pump therapy (-0.7% reduction vs. MDI + SMBG). Benefits were also greater among adults ≥25 years compared to younger patients, and among patients with higher adherence to CGM use (44,45). The results were heavily influenced by the STAR3 trial, and the JDRF study did not specifically compare outcomes between pump users and patients using MDI therapy. Severe hypoglycemia rates did not differ. However, the quality of most studies was limited due to small sample size, lack of blinding, and lack of sufficient data to compare hypoglycemia rates. Moreover, the meta-analyses were hampered by the inclusion of studies with obsolete technology or lack of consideration for the intended use of the device in the study (45,46). In another meta-analysis, studies that specifically enrolled patients at risk for hypoglycemia and used blinded CGM to assess it did show improvement in hypoglycemia (47).

A more recent meta-analysis of 21 studies published between 2011-2020 encompassing 2149 individuals with T1D revealed that CGM led to a significant reduction in A1c by 0.23% (p=0.0005), with larger treatment effect at higher baseline A1c (>8%), and no effect on severe hypoglycemia or DKA (48). However, this meta-analysis did not report CGM derived metrics such as TIR or TBR or clinically significant hypoglycemia. In a 2023 meta-analysis of 22 RCTs that included participants with T1D, there was an overall improvement in A1c, TIR and TBR (49). Reduction in A1c was limited to nonadjunctive devices but all devices resulted in improvement in TIR.

PATIENT REPORTED OUTCOMES

Generic Quality of life scores generally do not improve with rtCGM but treatment-specific measures, such as diabetes distress, hypoglycemic confidence, fear of hypoglycemia, and to a lesser extent, measures of convenience, efficacy and performance may be improved (33,50,51).

Evidence- Type 2 Diabetes

In patients with T2D, even in those who are not on insulin, rtCGM may act as a motivator and positive influence for patients to improve diet, activity, and medication-taking behaviors. The change in behavior can potentially lead to better glycemic control and weight loss (52). Moreover, periodic (every 3 months) short-term (14 day) use of rtCGM may be sufficient to achieve and maintain clinically relevant improvements in A1c in this population (53). Studies may be divided according to background therapies (insulin or non-insulin therapeutics).

STUDIES UTILIZING INSULIN THERAPIES AS BACKGROUND

In 2012, Vigersky et al. randomized 100 patients with T2D on basal insulin and anti-hyperglycemic agents to either a group that used rtCGM intermittently (2 weeks on, 1 week off) or a group that recorded SMBG four times per day for 12 weeks. At 12 weeks, they found a significantly greater reduction in A1c by 1.0% in the CGM group compared to 0.5% reduction in the SMBG group. The effect persisted up to the 40-week follow-up visit (54).

In 2017, Beck et al. conducted a randomized study to evaluate rtCGM use in 158 patients with T2D with mean A1c of 8.5% treated using MDI (55). Over a 24-week period, the A1c decreased to 7.7% in the rtCGM group compared to 8% in the group with usual care (mean difference -0.3%, p=0.022). RtCGM derived hypoglycemia and quality of life did not differ.

The Dexcom MOBILE study assessed patients with T2D on basal insulin randomly assigned to the Dexcom G6 or usual care for 8 months and reported a significant reduction in A1c, improved TIR and hypoglycemia (56). This was accomplished without an appreciable change in insulin or other medication use, indicating that CGM improves glucose levels by facilitating behavioral changes. Moreover, subsequent discontinuation of CGM for 6 months resulted in loss of about half of the improvement in TIR (57). The benefit was similar in older (≥65 years old) vs. younger adults (58).

In a 10-week study of 101 patients with T2D on MDI, patients randomized to isCGM (FreeStyle Libre) had greater A1c reduction (-0.82 vs -0.33%, p=0.005), found their treatment to be significantly more flexible and were more likely to recommend it to others compared to SMBG (59).

Among 141 adults with T2D treated with insulin or sulfonylureas and recent myocardial infarction, those randomized to isCGM had significantly less TBR (-80 minutes, 95% CI -118, -43 minutes) at 90 days, but marginal difference in A1c or TIR, and the intervention was reported to be cost-effective (60).

STUDIES UTILIZING ONLY NON-INSULIN THERAPIES AS BACKGROUND

In a randomized trial of 116 adults with T2D using non-insulin therapies, isCGM in combination with diabetes self-management education demonstrated superior A1c reduction at 16 weeks (treatment difference 0.3%, 95% CI 0 to

0.7%, p=0.048), larger increase in TIR (9.9%, p<0.01), and greater satisfaction compared to education (61).

In a 6-month randomized study of 72 adults with T2D not on insulin or sulfonylureas (baseline A1c 7.5-12%), use of CGM, either alone or with a food logging app, led to significant improvements in glycemic control without early medication changes. Time above 180 mg/dL dropped substantially in both groups by 3 months, with continued or sustained improvements at 6 months. TIR increased from 46% to 72%, A1c decreased by 1.3%, and participants lost an average of 7 pounds. The addition of food logging offered no clear advantage over CGM alone, suggesting that CGM feedback itself effectively drives lifestyle changes (62).

META-ANALYSES

A 2024 systematic review and meta-analysis which included 12 RCTs encompassing 1248 participants, demonstrated that CGM use was associated with a statistically significant A1c reduction of 0.31% (95% CI -0.47 to -0.21; p<0.00001). In addition, TIR increased by 6.4%, while TAR and TBR decreased by 5.9% and 0.7%, respectively. These benefits were observed in patients managed with insulin (± oral agents) and those using oral medications alone (63).

Another meta-analysis of 26 RCTs including 2783 patients with T2D found that both rt and isCGM resulted in modest but significant A1c reductions (-0.19% and -0.31%). RtCGM showed higher rates of device-related adverse events and lower satisfaction scores compared to isCGM, potentially due to alarm fatigue (64).

In a meta-analysis of 8 RCTs encompassing 541 persons with T2D on non-insulin therapies, CGM was associated with 0.37% reduction in A1c (95% CI -0.49, -0.24, p<0.00001), and increased TIR by 8.84% (95% CI 4.62, 13.06, p<0.0001) (65).

An umbrella review encompassing 15 systematic reviews with 81 studies and 23,657 participants reported a small but significant reduction in A1c (-0.23%, 95% CI -0.29, -0.18) with SMBG compared to no monitoring and an incremental greater A1c reduction with CGM compared to SMBG (-0.29, 95% -0.34, -0.24), and both isCGM and rtCGM were superior to SMBG (66).

PATIENT REPORTED OUTCOMES

While generic quality of life scores generally do not improve with rtCGM in T2D, treatment-specific psychosocial outcomes show consistent benefit. In insulin-using adults with T2D, rtCGM has been associated with significant improvements in hypoglycemic fear, confidence, and diabetes distress across both real-world and randomized settings (67–69). Notably, these benefits are not limited to insulin users; adults with T2D on oral medications transitioning to FreeStyle Libre 2 also demonstrated significant improvements in self-management capability, opportunity, and motivation (70).

Real World Outcomes

In a study of over 29,000 pediatric patients with T1D in the T1D Exchange Registry or the German/Austrian DPV Initiative, pediatric CGM use was associated with lower mean A1c regardless of insulin delivery modality (pump or injection) (71).

In a study of 106 UK hospitals incorporating 16,427 participants,1241 with repeated TIR data, improvements in TIR were associated with improvement in hypoglycemia unawareness and diabetes related distress. Moreover, TIR >70% was associated with reduced resource utilization (hospital admissions for hypoglycemia or hyperglycemia), paramedic visits and severe hypoglycemia (72).

In the Swedish National Diabetes Registry that included 14,372 adults with T1D, isCGM was associated with a small (0.11%, p<0.0001) reduction in A1c after 15-24 months and reduction in severe hypoglycemic episodes (OR 0.79, 95% CI 0.69-0.91) (73).

Using the French national claims database, a total of 74,011 patients with T1D or T2D initiated isCGM and over 98% persisted with the device at 12 months (74). Following initiation of the device, patients had a 39-49% reduction in hospitalizations for acute complications and a 32-40% reduction in diabetes-related coma. Moreover, the reduction in hospitalizations persisted after 2 years (75).

In Belgium, 1913 adults with T1D were studied before and after nationwide reimbursement of isCGM (76). Following the policy change, treatment satisfaction improved, there was a significant reduction in admissions for acute complications (severe hypoglycemia or ketoacidosis), and there were fewer absences from work.

Among 41,753 patients with insulin requiring diabetes in an integrated health care delivery system, 3806 patients initiated CGM, which was associated with a greater reduction in A1c (adjusted treatment difference 0.40%, $p<0.001$), fewer emergency department visits or hospitalizations for hypoglycemia (adjusted difference -2.7%, $p=0.001$), a reduction in number of outpatient visits and an increase in telephone visits (77). However, there was no difference in hospitalizations for hyperglycemia or ketoacidosis.

In a Medicare supplemental and commercial claims database study of 2463 patients with T2D on multiple injections of insulin/day, isCGM was associated with a reduction in acute diabetes events (HR 0.39, 95% CI 0.30-0.51) and all cause hospitalizations (HR 0.68, 95% CI 0.59-0.78) at 6 months compared to the 6 months prior to initiation (78).

The CORRIDA LIFE study was a 12-month, non-randomized real-world trial comparing rtCGM with isCGM among 191 adults with T1D. At 12 months, those using rtCGM had significantly lower A1c levels (7.1% vs. 7.7%, $p=0.0001$) and better TIR (67.5% vs. 57.8%, $p=0.0002$), with less time spent in hypoglycemia <70 mg/dL (4.3% vs. 6.4%, $p=0.003$) and <54 mg/dL (0.9% vs. 2.3%, $p<0.0001$) compared to isCGM users (79,80).

Real-world data complement the trial evidence by demonstrating that CGM benefits extend to routine clinical practice. Across national registries and claims databases, device persistence exceeded 98% at 12 months, and CGM initiation is consistently associated with reductions in hospitalizations for acute complications such as severe hypoglycemia, with some analyses showing persistent reductions beyond two years of follow-up (81–83). A1c improvements and reductions in emergency visits were similarly observed across pediatric and adult populations regardless of insulin delivery modality (84–86).

Recommendations

Patients should be adequately informed of the benefits and limitations of this technology, particularly with respect to the role for SMBG. Structured education programs should encompass concepts such as interpretation of and sharing data, customizing and troubleshooting alerts, addressing sensor adhesion and/or irritation, and understanding when to perform SMBG to confirm sensor readings. For those who are using MDI, education on carbohydrate counting and/or awareness and active insulin time (insulin on board) should be considered. Several expert groups have issued guidance on the use of rtCGM:

The 2023 Endocrine Society Clinical Practice Guideline: Management of individuals with Diabetes at high risk for hypoglycemia issued conditional recommendations for rtCGM in adults with T1D, CGM use in outpatients with T2D at high risk for hypoglycemia, and CGM initiation in select hospitalized patients at elevated hypoglycemia risk. This update reflects the growing role of CGM in both outpatient and inpatient settings (87).

The 2026 ADA Standards of Care recommend rtCGM to all patients with T1D on any insulin regimen, regardless of glycemic control and as soon as possible after the time of diagnosis (1). For adults with T2D, CGM is recommended for those using insulin or therapies known to cause significant hypoglycemia such as sulfonylureas. CGM may also be considered for individuals on non-insulin therapies to support individualized targets. Providers are advised to emphasize consistent use, with daily wear, and to ensure patients have continuous access to supplies to maximize benefit. Moreover, the guidelines recommend rtCGM rather than isCGM. In addition, persons using CGM should always have access to SMBG. Among pediatric patients, the ADA notes that CGM may reduce missed school days with regular usage (1).

Table 2.

ADA 2026 Recommendations for CGM

Group	***Recommendation***	***Level of Evidence***
Non-pregnant	At the time of diagnosis for all individuals using insulin therapy	A
	Non-insulin therapies that can cause hypoglycemia	C
	Any therapy where CGM helps in management Should be used as close to daily as possible	A
	The choice of device should be individualized based on patient centered factors	E
	People should have uninterrupted access to supplies to minimize gaps in monitoring	A
	Periodic or professional CGM can be helpful where continuous use is not possible	C
	Persons using CGM should always have access to SMBG	A
Diabetes and pregnancy	During pregnancy for individuals with T1D, to achieve glycemic goals (e.g., time in range and time above range)	A
	During pregnancy for individuals with T1D, to help achieve A1c goal	B
	May be beneficial for other types of diabetes in pregnancy.	E

A=Clear evidence from well-conducted, generalizable randomized controlled trials that
are adequately powered; B=Supportive evidence from well-conducted cohort studies; C= Supportive evidence from poorly controlled or uncontrolled studies; E=expert consensus. T1D=type 1 diabetes, T2D=type 2 diabetes, SMBG=self-monitored blood glucose.

The 2022 AACE Comprehensive Care Plan Guideline and 2023 AACE T2D Algorithm strongly advocate CGM use in all persons with T1D regardless of insulin delivery method, and in those with T2D treated with insulin or at high risk for hypoglycemia respectively. These guidelines also formally incorporate CGM-derived metrics as treatment targets alongside A1c, and offer more specific guidance on CGM modality selection based on hypoglycemia risk and clinical context (88,89).

In 2017, the Advanced Technologies & Treatments for Diabetes (ATTD) Congress organized an international consensus panel to analyze the existing literature and to provide guidance for utilizing, interpreting, and reporting CGM data (90). This was updated in 2019 (Table 3, 4). These recommendations are supported by data from the DCCT demonstrating that a 10% reduction in TIR derived from 7-point SMBG profiles is associated with a 40% reduction in risk of albuminuria and a 64% reduction in risk of incident or progressive retinopathy (91). In 2023, an updated consensus statement was published to standardize CGM reporting in clinical trials, mitigating variability across CGM systems (9).

Table 3.

CGM-Based Targets for Different Diabetes Populations

Glucose Range	**%Time in Range**	
	Non-Pregnant Patients	
	Type 1 and Type 2 Diabetes	**Older/High Risk Diabetes**
>250 mg/dL (13.9 mmol/L)*	<5%	<10%
<180 mg/dL (10 mmol/L)	<25%	<50%
70-180 mg/dL (3.9-10 mmol/L)	>70%	>50%
<70 mg/dL (3.9 mmol/L)**	<4%	<1%
<54 mg/dL (3.0 mmol/L)	<1%	
	Pregnant Patients	
	Type 1 Diabetes	**Gestational and Type 2 Diabetes** #
>140 mg/dL (7.8 mmol/L)	<25%	-
63-140 mg/dL (3.5-7.8 mmol/L)	>70%	-
<65 mg/dL (3.5 mmol/L)	<4%	-
<54 mg/dL (3.0 mmol/L)	<1%	-

* Includes time >180 mg/dl, **Includes time <54 mg/dl, #Insufficient data.

Also, in 2017, the ADA and the EASD published a joint statement providing recommendations for systematic improvements in clinical use and regulatory handling of CGM devices (92). Freckmann et al. conducted a scoping review identifying major inconsistencies in CGM performance studies, particularly in comparator methods, study design, and reported metrics (93). Accuracy varied significantly based on reference method, with up to 8% MARD differences between venous and capillary glucose. In a recent consensus statement, the IFCC released a validated framework for performance validation of CGM systems that would improve transparency, reduce variability in methods for assessing performance, but not resolve differences related to varying methods of manufacturer calibration (94). In addition, the IFCC established guidelines for clinical assessment, including requirements for study design, distribution of comparator measures, minimum requirements for accuracy (point accuracy, trend accuracy, sensor specific), and requirements for reporting performance (Table 5) (95).

Table 4.

Summary of ATTD Recommendations for CGM

Limitations of A1C	CGM should be utilized when there is a discrepancy in A1C and other measures of glucose control CGM should be utilized to assess hypoglycemia and glucose variability
Guiding management and assessing outcomes	CGM should be considered for patients with T1D and insulin treated T2D who are not achieving targets or those with hypoglycemia All patients should receive training regarding how to interpret and respond to their data, utilizing standardized programs with follow-up
Performance	No accepted standard exists for CGM system performance. However, a mean absolute relative difference ≤10% provides little additional benefit for insulin dosing.
Definition and assessment of hypoglycemia	Clinical classification Level 1: 54-70 mg/dL with or without symptoms

	Level 2: <54 mg/dL with or without symptoms (clinically significant) Level 3: cognitive impairment requiring external assistance for recovery Quantification using CGM % of values or time below a given threshold (54 or 70 mg/dL) Number of events (defined as CGM readings persistently below threshold for at least 15 min. with recovery defined as persistent readings over the threshold for at least 15 min.) over a given reporting period)
Glycemic Variability	Coefficient of Variation should be the primary measure
Time in Range	The % time in hyperglycemia, hypoglycemia, and target range should be reported.
CGM Metrics	Standardized reporting using the AGP and integration into electronic health records is recommended. Key metrics: Number of days worn (14 days recommended) % time CGM is active (70% of data from 14 days recommended) Mean glucose Glucose Management Indicator Glycemic variability (%CV, target <36%) % Time in Range (TIR): 70-180 mg/dL (3.9-10.0 mmol/L) %Time Above Range (TAR): 181-250 mg/dL (10.1-13.9 mmol/L), and >250 mg/dL (>13.9 mmol/L) %Time Below Range (TBR): 54-69 mg/dL (3.0-3.8 mmol/L), and <54 mg/dL (3.0 mmol/L)

CGM=continuous glucose monitor, T1D=type 1 diabetes, T2D=type 2 diabetes, SMBG=self-monitored blood glucose, AGP=ambulatory glucose profile.

Table 5.

Summary of IFCC Proposed Requirements for reporting CGM Performance

Accuracy	Point accuracy (MARD, bias, stratified by glucose range) Trend accuracy, stratified by glucose range Sensor-specific accuracy (% of values +/-20% Clinical point accuracy (DTS Error Grid Analysis) Agreement when outside measurement range
Stability	Accuracy by subperiod of wear (beginning, middle, end)
Alert Reliability	True/false alert and detection rate for hypo/hyperglycemic thresholds Trend alert reliability
Technical reliability	Sensor survival/failure % of recovered CGM data/gaps Time lag
Safety	Adverse device effects, severity and type

IFCC=International Federation of Clinical Chemistry and Laboratory Medicine, MARD=mean absolute relative difference, DTS=Diabetes Technology Society, CGM=continuous glucose monitor.

Hospital Use

CGM is not currently approved for use in the hospital setting. However, during the COVID-19 pandemic, the FDA announced that it would not object to their use in an effort to support reduction in use of personal protective equipment and risk of exposures to staff. Thus, there has been increasing interest in their use. Moreover, as increasing numbers of patients are using these devices in the ambulatory setting, they want to continue their use in the hospital. (96) Recent randomized trials support use of CGM on the hospital wards, where it has been shown to be safe, and may reduce the frequency of hypoglycemia (97,98). A 2025 meta-analysis of six RCTs comparing hybrid CGM plus point of care (POC) protocols with standard POC monitoring demonstrated significant improvements in glycemic outcomes, including a 7.2% increase in TIR (70-180 mg/dL, $p<0.00001$), a 3.7% reduction in time >250 mg/dL ($p=0.003$), and modest but clinically meaningful reductions in time <70 mg/dL (-1.23%) and <54 mg/dL (-0.95%). The magnitude of benefit appears to depend on how CGM data are incorporated into structured insulin titration protocols, with larger improvements observed when CGM guides algorithmic dosing adjustments rather than just providing additional monitoring (99). In addition, observational data show that CGM detects hypoglycemia more reliably than routine POC testing, identifying episodes in 56% of patients compared to just 14% by POC, underscoring the value of continuous data measurements in the hospital (100).

In the ICU, there is concern that CGM may be less accurate due to factors such as edema, hypoperfusion, and acidosis but studies suggest use of CGM in conjunction with periodic POC blood glucose within well-established protocols is safe and may reduce the need for POC BG (101). Observational data suggest that calibration in response to CGM-POC discordance can significantly improve sensor accuracy and may serve as a practical strategy to support safe inpatient CGM use (102).

In 2020, the Diabetes Technology Society sponsored a panel of experts in inpatient diabetes management to review the evidence for use of CGM in the hospital (103). The panel agreed that CGM had the potential to improve clinical outcomes, but at the time, limited data demonstrating accuracy and clinical utility was available, and there is a lack of decision support systems, including infrastructure for communicating results to care teams and to the electronic medical record. Owing to the post-pandemic growth of inpatient CGM use, a 2024 international consensus statement, endorsed by the ADA and the EASD, published updated recommendations that provided guidance on structured integration of CGM into hospital systems. The expert panel issued several Good Practice Points to guide safe and effective CGM use in both general ward and ICU settings:

> Protocol Development: Hospitals should implement standardized hybrid protocols that combine CGM with scheduled POC glucose checks. These protocols should define what constitutes acceptable agreement between CGM and POC values, typically within 20%, and outline when confirmatory testing is needed, such as for glucose values <70 mg/dL or when there is clinical concern about accuracy.
>
> Patient-owned devices: Patients already using personal CGM devices should be allowed to continue them in the hospital if they’re able to manage their device safely. Institutions should have clear policies in place that address safety, including continued POC testing, involvement of endocrinology or diabetes teams, and signed waivers that explain the risks and limitations of inpatient CGM use.
>
> Operational Considerations: Successful CGM use in the hospital requires clear operational workflows. This includes creating process maps, tailored order sets, and defining team roles for sensor placement, data review, and treatment response. Hospitals should ensure CGM data is clearly labeled in the electronic medical record and distinguishable from POC glucose. Integration of unique device identifiers is encouraged.
>
> Staff Training and Safety: All clinical staff involved with CGM should receive training on interpreting CGM data, troubleshooting issues, and identifying sensor errors such as compression artifacts that can cause falsely low readings. Sensors should be removed before MRI or CT imaging, following manufacturer guidelines.
>
> Alert and Alarm Management: Hypoglycemia alerts should be customized to the patient’s clinical status and institutional targets. Protocols should support timely responses while minimizing alarm fatigue, particularly in

environments already burdened by frequent alerts.

Limitations of Use

It should be emphasized that most prospective RCTs enroll highly motivated patients and have methodological limitations. Many studies are sponsored by industry, have short durations and use open-label designs, with significant heterogeneity in reporting outcomes. In the real-world setting, there are concerns about limited resources for training, and less motivated patients may be overwhelmed with the additional data, particularly where complex algorithms are required.

Despite these challenges, CGM adoption has grown substantially over time. In the T1D Exchange Registry, CGM use increased from 7% in 2010-2012 to 30% in 2016-2018, and rose more than 10-fold in children (104)/ A1c levels were lower in CGM users compared to nonusers. In a Breakthrough T1D (formerly JDRF) study, any device use (CGM or insulin pump or both) climbed from 19.7% to 77.4% from 2014 to 2023. In particular, CGM use increased after 2017, when Medicare broadened coverage (105). However, there was a lag between increases in any use of CGM and consistent use, requiring further study. It is likely that adherence has improved as technologies evolved in terms of accuracy, usability (no need for calibration, smaller size), and broader coverage.

Nevertheless, several barriers contribute to suboptimal CGM uptake and discontinuation. More than half of respondents in one study cited cost or insurance coverage as a significant barrier to use (106). Moreover, disparities in prescribing patterns and implicit bias have been described (107–110). Modifiable reasons for avoiding use include the hassle of devices (47%) and aversion to having a device attached to the body (35%). Skin reactions and/or difficulty with adhesion are well known and are an important cause of discontinuation (111). Methods of addressing this barrier (such as use of barriers, overlay patches, or topical antihistamines and corticosteroids have been described but additional research is needed (112). Meighan et al found that the most common reasons for CGM attrition were attributable to device adhesion, dislike of having a device on the body, pain with device use, and system mistrust due to inaccurate readings (113).

The 2023 International Consensus Statement on CGM and Metrics for Clinical Trials noted that most accuracy studies have predominantly enrolled White participants with few Black or Asian participants, limiting generalizability (9). A 2025 study by Alkabbani et al. using Medicare and commercial claims data found that compared with White patients, Hispanic individuals have 0.44 odds (Medicare) and 0.81 odds (commercial insurance) of receiving CGM, while Black patients have 0.71 odds (Medicare) and 0.89 odds (commercial insurance) of receiving CGM.(114) Socioeconomic factors compound these disparities, as older age, lower socioeconomic status, public insurance, and residing in deprived areas are all associated with lower CGM uptake (114,115).

Other limitations include possible interference with acetaminophen, ascorbate, and other active agents in glucose-oxidase-based electrochemical sensors. They are also dependent on both the sensitivity and specificity on the enzyme availability on the electrode surface. Other identified interfering substances include hydroxyurea, which causes temporary falsely elevated sensor glucose values, and mannitol and sorbitol when administered intravenously or as a component of peritoneal dialysis solution, which may cause falsely elevated readings (1). Importantly, these interferences may be device specific. There are well known delay artifacts due to the time lag between glucose concentration in the interstitial fluid and blood glucose and are magnified during rapid changes in glucose. This delay, typically between 5 and 10 minutes, is not crucial to analyzing retrospective data, but can be critical when CGM is indicated for real-time decision making (116).

Daily Use

Patients must be aware that sensor readings can deviate from actual blood glucose measurements, particularly during rapid glucose changes such as that which occurs post-meal or during exercise. Calibration, where necessary, should not be performed when trend arrows indicate rapid swings in glucose. While systems are becoming more reliable, patients may need to verify sensor readings before taking action such as meal boluses or treatment of hypoglycemia depending on the device, even if a device is approved for nonadjunctive use. Alarm thresholds should be set in order to maximize patient adherence, keeping in mind that the sensitivity for detecting hypoglycemia decreases as the threshold is reduced below 70 mg/dL (117). Conversely, specificity improves to a much smaller degree at lower thresholds, and thus false alarms may not be reduced substantially.

Several algorithms have been published that provide specific guidance to patients for responding to trend arrows and alarms and are summarized below. All algorithms are complex (118–122). Therefore, they should only be implemented in patients who have demonstrated an understanding of CGM technology, including lag times between CGM and SMBG, calibration procedures, alerts and trend arrows, as well as understanding of insulin action time and the risks of insulin stacking. In one small study, trend arrows were accurate approximately 79% of the time outside of mealtime windows (30 minutes before and 120 minutes after carbohydrate intake) but this dropped to ~60% within mealtime windows (123). Thus, algorithms are not intended for use post-meal. The use of AID systems (discussed later) increases safety and efficacy and reduces the complexity of the trend arrow approach. Moreover, future artificial intelligence informed platforms could provide more contextual information beyond glucose levels and long-range predictive power to ultimately reduce diabetes burden(124,125).

A consensus statement facilitated by the Endocrine Society provides expert guidance on the use of trend arrows for making treatment decisions (126). The guidance recommends adjustment of boluses pre-meal and no sooner than 4 hours post-meal in 0.5-unit increments based upon the trend arrow and the patient's sensitivity. The statement recommends no additional treatment within 2 hours of a previous meal bolus, and correction bolus using the bolus calculator or usual correction dose only in the 4 hours after a meal. Similar expert guidance has been developed for the FreeStyle Libre system (127).

A more recent adaptation of the Endocrine Society guidance incorporated pre-meal glucose levels in addition to the insulin (128), and a small randomized study demonstrated it was more effective than the incorporation of insulin sensitivity alone, though this was limited to insulin pump and not MDI therapy (129).

A multi-center trial evaluated a CGM-informed bolus calculator (CIBC) with automatic insulin dose adjustment based on CGM trend, used with the Omnipod 5 system in 25 patients. Compared to a standard bolus calculator, the CIBC resulted in significantly fewer sensor readings below 70 mg/dL in the 4 hours following a bolus, while overall TIR remained unchanged, demonstrating that automated trend-based bolus adjustment can reduce postprandial hypoglycemia (130).

Overview of Stand-alone Personal rtCGM and isCGM Systems

The first rtCGM (Guardian, Medtronic[R]) was approved in 2004. Since then, additional models and other devices have entered the market, and accuracy and patient satisfaction have improved. Several personal CGMs have been approved by the FDA for use in the United States or carry CE marking for use in Europe and are currently on the market (Table 6). For a full review of regulatory requirements for glucose monitoring devices, the reader is referred to one of several excellent reviews and consensus statements (94,95).

Table 6.

Comparison of Subcutaneous Continuous Glucose Monitoring Devices

	Calibration required[1]	Real-time alerts	Sensor Life (days)	Warm-up (hrs)	Remove for MRI, CT diathermy[2]	Acetaminophen interference[3]
Dexcom G6	N	Y	10	2	Y	N

	Calibration required[1]	Real-time alerts	Sensor Life (days)	Warm-up (hrs)	Remove for MRI, CT diathermy[2]	Acetaminophen interference[3]
Dexcom G7	N	Y	10	0.5	Y	N
Dexcom G7 15-day	N	Y	15.5	0.5	Y	N
MiniMed Guardian 4	N	Y	7	2	Y	Y
MiniMed Simplera	N	Y	7	2	Y	Y
Minimed Instinct	N	Y	15	1	Y	N
FreeStyle Libre 2+	N	Y	15	1	N	N
FreeStyle Libre 3+	N	Y	15	1	N	N
Eversense 365 (surgical implant)	Y	Y	365	24	Y (remove transmitter)	N

1 All devices are labeled for nonadjunctive use. Calibration required for the Eversense 365 during initiation (periodically in the first 14 days and weekly thereafter).

2 Freestyle Libre sensors have FDA approval for continued wear during imaging. Sensor accuracy may be compromised during MRI but recovers within 1 hour. Scanning with 3T scanners should be limited to 12 minutes with cooling period of 2 minutes between scans for the torso and up to 1 hour for other areas or with 1.5T scanners (131–133). Preliminary data demonstrates retained CGM accuracy during X-ray and CT (133).

3 No interference with Dexcom G6 or G7 at maximum recommended doses of 1000 mg every 6 hours.

DEXCOM G6

The Dexcom CGM utilizes a glucose oxidase sensor at the tip of a wire that is implanted in the subcutaneous space. The data are transmitted wirelessly and are displayed on a separate receiver (personal smartphone or device specific receiver). The Dexcom G6 has a sensor life of 10 days, a 2-hour warm-up period, does not require calibrations, and minimizes interference by acetaminophen. The Dexcom G6 is approved for individuals ≥2 years of age and for nonadjunctive use. Adults wear the sensor on the abdomen, while youth ages 2-17 years may also use the upper buttocks. G6 is also associated with a smartphone app that allows the patient to log activity, set reminders or alarms, and physically see their glucose levels and trends throughout their time wearing the device. G6 is compatible with both iOS and Android devices. This system features real-time alerts including an “Urgent Low Soon” predictive alert for impending hypoglycemia. The G6 was the first CGM to receive FDA clearance as an integrated CGM in 2018. The Dexcom CLARITY Diabetes Management Software organizes and presents the patient’s glucose data. Dexcom SHARE app allows users to share data with up to 10 other individuals. The Dexcom G6 is being discontinued in 2026.

DEXCOM G7

The Dexcom G7 features a 60% smaller size (the size of 3 stacked quarters) vs. the G6, is fully disposable, and has a shorter 30-minute warm up time and a 12-hour grace period to replace completed sensors. The sensor life is 10 days with a 12-hour grace period. The overall MARD was reported to be 8.2% with the arm and 9.1% on the abdomen

(20). Consistent with previous studies, accuracy is lower at lower sensor glucose, higher glucose rate of change, and on day one of wear. G7 is approved for individuals ≥2 years of age, including use in pregnancy. Individuals ≥7 years can wear the sensor on the back of the upper arm, while those ages 2-6 years can use the upper arm or upper buttocks. The G7 is factory calibrated and approved for nonadjunctive use and is also cleared as an integrated CGM. Optional calibrations may be performed. The G7 app is available on both iOS and Android devices and integrates with Apple Watch. New features include delayed high glucose alerts (15 minutes to 6 hours), quiet mode vibrate, quiet mode silence all (up to 6 hours), and the ability to store 24 hours of data for backfill when Bluetooth connection is restored. The Urgent Low Soon predictive alert warns of impending hypoglycemia. Data can be shared with followers via Dexcom SHARE and managed through Dexcom CLARITY software via the Dexcom G7 app.

DEXCOM G7 15-DAY

The Dexcom G7 15-day sensor maintains the same fully disposable design as the standard G7 with the same 30-minute warm up time but extends the wear duration from 10 days to 15 days. The 15-day version demonstrated an overall MARD of 8.0% across the extended wear period, with accuracy maintained throughout all 15 days of use, and can be used nonadjunctively (134). The system retains all other G7 features, including the 12-hour grace period, smartphone connectivity, data sharing capabilities through Dexcom SHARE, and integration with Dexcom CLARITY software. This device exceeded FDA CGM performance goals and can be used as an integrated CGM.

MINIMED GUARDIAN 4 SENSOR

The Guardian Sensor 4 (GS4) is the successor to the Guardian Sensor 3. The sensor transmits glucose data every 5 minutes via Bluetooth to compatible devices, including the MiniMed 780G insulin pump for use in automated insulin delivery and to a smartphone via the MiniMed Mobile app. The GS4 is factory calibrated and approved for nonadjunctive use, though users may still perform optional calibrations if desired. The sensor has a 7-day wear time and a 2-hour warm-up period.

MINIMED SIMPLERA

The Simplera is Medtronic's first disposable, all-in-one CGM sensor, combining sensor and transmitter into a single unit and eliminating the need for a separate reusable transmitter as required by the Guardian 4 (135). The sensor is smaller and flatter than the Guardian 4, does not require overtape, and uses a two-step insertion process worn on the back of the upper arm. The sensor has an 8-day wear time, a 2-hour warm-up period, and is factory calibrated, though optional calibrations are advised when sensor readings do not match symptoms. The Simplera is approved for individuals ages 7 and older with T1D and adults ages 18 and older with insulin-requiring T2D. Data are transmitted every 5 minutes via Bluetooth and are viewable on compatible iOS and Android devices including Apple Watch. The standalone Simplera received FDA approval in August 2024 and integrates with the Medtronic InPen smart insulin pen for MDI patients. A variant, the Simplera Sync, was separately approved in April 2025 for exclusive use with the MiniMed 780G AID system and is discussed further in the AID section.

In a head-to-head accuracy study comparing the Simplera, Dexcom G7, and FreeStyle Libre 3 against Yellow Springs Instrument venous reference in 24 adults with T1D, the Simplera demonstrated a MARD of 11.6%, similar to the FreeStyle Libre 3 (11.6%) and Dexcom G7 (12.0%) (136). However, accuracy against blood glucose was lower for the Simplera (MARD 16.6%). The Simplera showed better accuracy in the hypoglycemic range, while the G7 and Libre 3 performed better in normoglycemic and hyperglycemic ranges.

MINIMED INSTINCT

The Instinct sensor is a disposable, all-in-one rtCGM designed by Abbott for use with the MiniMed 780G system (137). FDA clearance for integration with the MiniMed 780G was granted in September 2025. The Instinct is worn on the back of the upper arm using a one-step applicator and has a 15-day wear time with a 1-hour warm-up period. It is approved for use in individuals ages 2 and older. It is factory calibrated and does not require fingersticks, though blood glucose confirmation is recommended when sensor readings do not match symptoms or during the first 12 hours of wear. Readings are transmitted every minute to the connected app. The Instinct sensor is exclusively compatible with MiniMed systems and is not interoperable with other AID platforms and cannot be used as a stand-alone CGM.

FREESTYLE LIBRE 2 PLUS

The FreeStyle Libre 2 Plus is a modified version of the FreeStyle Libre 2. This system includes an extended 15-day sensor life (compared to 14 days in the FreeStyle Libre 2), is approved for individuals ages 2 and older and is cleared for use in pregnancy. It is factory calibrated and is approved for nonadjunctive use. It features a 1-hour warm-up time and is compatible with both iOS and Android devices. Data are managed through the FreeStyle LibreLink app and LibreView cloud platform. The Libre 2 Plus has also been cleared as an integrated CGM.

FREESTYLE LIBRE 3 PLUS

The FreeStyle Libre 3 Plus is the smallest sensor available. It offers a 15-day sensor life with a 1-hour warm-up period. It is factory calibrated with a Bluetooth range of 33 feet. It is approved for ages 2 years and older including pregnancy. Cleared as an iCGM for AID system integration, it demonstrated a MARD of 7.5–11.6% in accuracy studies, with performance comparable to the Dexcom G7 (136). Data are managed via the FreeStyle Libre 3 app and LibreView platform, with a dedicated display device available for Medicare coverage (138).

EVERSENSE 365

The Eversense 365 is the first and only fully implantable CGM, FDA-cleared in 2024, with a sensor life of 365 days. The sensor is implanted under the skin of the upper arm by a healthcare provider in an outpatient procedure using local anesthesia. It uses fluorescence-based glucose-sensing technology instead of the electrochemical method found in other CGMs. A removable, rechargeable transmitter worn over the implant sends real-time data to the Eversense app (compatible with both iOS and Android) and can be taken off for swimming, showering, or contact sports without interrupting sensor function. The system requires roughly one calibration per week after the warm-up period and features on-body vibration alerts. In the ENHANCE pivotal study, it achieved an overall MARD of 8.8%, 90% sensor survival at 365 days, and confirmed alert detection rates of 96.6% at 70 mg/dL and 97.9% at 180 mg/dL (139). It is approved for nonadjunctive use and indicated for adults ≥18 years. Notably, interference has been reported with tetracycline and mannitol, though not with acetaminophen or ascorbic acid. Adverse events were uncommon, and no device- or procedure-related serious adverse events were reported in ENHANCE. Data are managed through the Eversense Data Management System. The implantable design may be particularly useful for patients with privacy concerns, physical disability, needle phobia, allergies or other difficulty with adhesion, or activities or professions that may be barriers to external wear (140).

OVER-THE-COUNTER DEVICES

Recently, over-the-counter devices have become available and are marketed to patients with diabetes on non-insulin therapies or even to persons without diabetes (prediabetes, obesity, or even among performance athletes) in order to inform or reinforce diet and physical activity. Moreover, studies demonstrate that CGM can detect differences in dietary patterns and physical activity in individuals without diabetes (141). Of particular interest is that glycemic fluctuations are associated with oxidative stress, markers of inflammation, endothelial dysfunction, and autonomic dysfunction, suggesting a possible link with cardiovascular disease (141). Moreover, there is interest in use of CGM to predict diabetes and offering precision nutritional approaches to address specific “glucotypes” (142,143). However, it remains to be established whether CGM can be recommended more broadly as an effective weight loss intervention and whether the use of CGM impacts other important health outcomes in persons without diabetes. Over-the-counter devices may be useful for individuals with diabetes who are not using therapies that cause hypoglycemia since these devices do not have audible alerts. Audible alerts can cause distress or prompt inappropriate treatment in individuals who are not at risk for hypoglycemia. These devices include the 15-day Dexcom Stelo and the 15-day Abbott Libre Rio. The Abbott Lingo is designed specifically for wellness with an accompanying lifestyle coaching app.

STEPS TOWARDS AN ARTIFICIAL PANCREAS

Historically, rtCGM technology has operated completely independently of insulin delivery. By combining continuous basal insulin delivery during fasting periods with discrete bolus doses of insulin at mealtimes, insulin delivery can be crafted to mimic the natural pattern of pancreatic insulin release. AID systems seek to incorporate CGM data to adjust insulin delivery further. An AID consists of 1) a CGM; 2) a continuous insulin delivery system; and 3) a control processor to link the insulin delivery to the glucose level. Limitations to full implementation of an artificial pancreas include sensor accuracy and lag time, inadequate onset and offset of currently available rapid acting insulin analogs, meal challenges, and changes in insulin sensitivity due to circadian rhythms, exercise, menstrual cycles, and intercurrent illness (144). However, even incremental advances over time have demonstrated improved glucose control without increasing the complexity of decision-making on the part of the patient. These include:

Low glucose (threshold) suspend: the insulin pump suspends when the glucose decreases below a pre-set value.

Suspend before low: insulin pump suspends when hypoglycemia is predicted.

Hybrid closed loop: insulin delivery increases or decreases based upon the sensor glucose value, but meal boluses are still required.

Closed loop control: fully closed loop delivery without the need for meal boluses.

Dual hormone systems: these are hybrid closed loop or closed loop control systems that utilize glucagon or other peptides (such as amylin) in an effort to more closely mimic the physiology of the endocrine pancreas.

Most currently available systems now provide hybrid closed loop delivery. The long-term safety, efficacy, cost, and cost-effectiveness of an artificial pancreas are still largely unknown at this time. Of these systems, Tidepool Loop which was originally developed through crowd-sourced software received FDA approval (145). The reader is referred to one of several reviews cited, as a detailed review of AID technology is beyond the scope of this chapter (146,147).

Hybrid Closed Loop (HCL)

HCL refers to sensor glucose driven automatic adjustment of insulin delivery with or without additional auto boluses and still requires the patient to bolus for meals. Advanced hybrid closed-loop (AHCL) systems including MiniMed 780G and Control-IQ represent the current generation that adjusts basal insulin and delivers automatic correction boluses, though user-controlled prandial insulin administration is still required. In a recent consensus statement (148). an ideal candidate for automated insulin delivery systems:

Is technically capable of managing a pump, has basic carbohydrate counting skills or carbohydrate awareness, and are able to implement a back-up plan (including the use of manual injections).

Has realistic expectations of system capabilities. In particular, several situations that are unique to HCL are worth emphasizing:

Bolusing: Pre-bolusing approximately 15 minutes prior to meals with use of rapid acting insulins or immediately premeal with ultra-rapid insulins is critical to maintain TIR. In many systems, delayed boluses not only cause early post meal hyperglycemia but also precipitate delayed hypoglycemia as the system has already begun to augment insulin delivery in response to hyperglycemia.

Exercise management: Similarly, carbohydrate loading prior to exercise while using HCL systems will only stimulate insulin delivery and thus it is recommended that users implement other means for management such as setting a higher target, typically with a designated exercise mode, or exiting to manual mode with temporary basal insulin reduction or temporary suspension of the pump.

Hypoglycemia management: HCL users typically need fewer carbohydrates (about half) to manage hypoglycemia since the pump has generally already suspended insulin delivery based upon glucose trends.

Has adequate support, including diabetes education, insurance coverage, and caregiver or other social support where relevant.

Has the ability to transmit data to the healthcare team.

Is mentally and psychologically able to implement AID systems.

On the other hand, there does not appear to be an ideal threshold A1c for determining candidates for HCL therapy, as those with lower A1c may benefit by reducing TBR and those with higher A1c benefit from reductions in hyperglycemia without the perceived risk of ketoacidosis traditionally attributed to initiation of insulin pump therapy (149). Thus, less ideal candidates may obtain the greatest benefit in terms of achieving glycemic targets.

Most studies have enrolled patients with T1D. More data are needed for special populations including T2D, especially those with very high insulin requirements, pregnancy, acute illness, steroid use, renal dysfunction, and persons in assisted living facilities (148). Recent data including the ORACL randomized crossover study comparing closed-loop to sensor-augmented pump therapy in patients ages 65 and older with T1D showed that closed-loop improved TIR by 6.2 percentage points, with the greatest benefit overnight and a meaningful reduction in nocturnal hypoglycemia, a particularly important finding in this population (150).

As with CGM, it is important to evaluate these systems in the real-world setting, where user experience can differ from that of the highly controlled and supportive research environment. Cost and insurance hassles, as well as user wear issues are the most commonly reported barriers to use of any diabetes related device. These barriers contribute to diabetes distress and depressive symptoms which can impede self-management behaviors (151). HCL systems improve glucose control but may also introduce additional alarms or alerts which are needed for safety (such as threshold alerts for hyperglycemia or hypoglycemia or HCL mode exits due to insulin delivery exceeding the system's guardrail) or ongoing functionality such as calibration of the CGM (152). Newer systems have attempted to address many of these barriers through improved algorithms or other features (Table 7). A systematic review found that despite the complexity, HCL systems led to stable or improved diabetes distress, fear, quality of life and improved sleep (153). Real-world data from the National Health Service in England pilot study demonstrated that 95% of users (540 of 570 participants) stated that HCL had a positive impact on quality of life (154,155).

Devices differ considerably with respect to algorithms used for insulin adjustment and a number of other features (Table 7) (4,156,157). There are few head-to-head studies comparing the efficacy and safety of available HCL systems. In a recent meta-analysis, all systems except iLet were associated with mean TIR ≥65% after controlling for baseline A1C, though MiniMed 780G had the highest TIR in cohorts ≥age 18. Moreover, the ADA recommends choice of AID system based upon clinical needs and preferences (1). Details of select systems are presented next.

MEDTRONIC 670G

The first system to gain FDA approval is the Medtronic 670G, which adjusts basal insulin delivery every 5 minutes when in auto mode. This system utilizes the Guardian sensor 3, which offers enhanced sensor accuracy, with an overall MARD of 9.64% (158). The system was associated with a reduction in A1c from 7.4 to 6.9% and there were trends in improvement of TIR and hypoglycemia in a non-randomized study of 124 patients with T1D (159). A subsequent randomized trial of 151 adults and children demonstrated a significant reduction in A1c and TBR compared to insulin pump without CGM.(160) A 6-month RCT of 120 adults demonstrated TIR increasing from 55% to 70% with no change in the control group and lower A1c and improved diabetes-specific well-being with HCL (150). A pivotal RCT versus CSII without CGM confirmed meaningful A1c reductions in both higher and lower baseline A1c subgroups, with reduced TBR and no severe hypoglycemia in the HCL group (160). A large international multicenter RCT further established superiority of 670G/770G over MDI across subgroups (161). A systematic review of 30 studies in youth supported the system's safety and efficacy with TIR reaching up to 73%, though operational demands were consistently perceived as burdensome, contributing to declining auto mode use over time (162). Long-term real-world use remains a concern. In a 1-year prospective observational study of 84 patients, 28% stopped using auto mode by 3 months, and 33% discontinued by 12 months (163). The most common reasons for discontinuation included sensor issues (62%), difficulty obtaining supplies (12%), fear of hypoglycemia (12%), preference for injections (8%) or sports (8%). In a study of 92 youth, 30% discontinued HCL, typically between 3 and 6 months after initiation, due to issues such as difficulty with calibrations, alarms, and extra time needed for operation (164).

MEDTRONIC ADVANCED HYBRID CLOSED LOOP (AHCL) SYSTEM (780G)

The MiniMed 780G builds on the 670G/770G platform with meaningful upgrades including a glucose target adjustable as low as 100 mg/dL, automated correction boluses every 5 minutes, compatibility with the factory calibrated Guardian 4 sensor, remote software updating, and smartphone data viewing (165). It received FDA clearance for individuals with T1D ≥7 years and is also approved for T2D. The pivotal single-arm study of 157 adolescents and adults showed nearly 95% time in automated mode with only 1.2 exits per week and improvements across A1C, TIR, and TBR (166). The ADAPT trial, the first RCT of the 780G confirmed a 1.42% A1c reduction versus MDI plus isCGM alongside improved treatment satisfaction and reduced fear of hypoglycemia (166). By comparison, in a randomized study of 41 participants with T1D who were naïve to both CGM and insulin pump technologies, AHCL also resulted in improvements in TBR (167). In a real-world study of 3211 youth (age <15 years) and 8874 individuals >age 15 years, AHCL demonstrated >90% treatment persistence over 6 months (168). The AHCL system was reported to have better glucose monitoring treatment satisfaction but similar diabetes distress, technology attitudes, and fear of hypoglycemia compared to the 670G system (169).

Compatibility with the fully disposable and self-contained 7-day Simplera Sync sensor and Extended Wear Infusion Sets has further improved automation and wear time in real-world use. Finally, the Instinct sensor was developed through a partnership with Abbott to operate with the 780G system and provides an even smaller size with 1 hour warm up period and 15-day wear time (170). The FDA cleared the MiniMed Flex pump in 2026, which is screenless and controlled via smartphone app but uses the 780G algorithm.

TANDEM CONTROL-IQ

The Tandem t:slim X2 with Control-IQ pairs a model predictive control algorithm with the calibration-free Dexcom G6 or G7 or FreeStyle Libre 3 Plus sensors, automating basal adjustments every 5 minutes, delivering correction boluses, and intensifying overnight control to target 100-120 mg/dL (171,172). This system is FDA approved for T1D ages 2+ and is also approved for patients with T2D (173). In a 6 month trial of 168 patients (age 14-71) randomized 2:1 to HCL vs. sensor-augmented pump alone, the TIR was increased by 11% more in the HCL group compared to sensor-augmented pump ($p<0.001$), with improvements in hypoglycemia, mean glucose and A1c (171).

A pooled meta-analysis of three RCTs (n=369, ages 2–72) confirmed a consistent +11.5% TIR benefit across race, socioeconomic status, pump experience, and baseline glycemic control (174). Control-IQ has also been studied in adults at high risk for hypoglycemia, reducing TBR while improving TIR (175) and in pregnancy, where the CIRCUIT RCT showed a 15 percentage point TIR improvement over standard care (176); although not FDA-cleared for this use. Real-world data have been reassuring, with one study including 1435 persons with T1D showing reduced impact of diabetes on life, improved device related treatment satisfaction, and improved emotional well-being (177). A Belgian prospective study showed sustained gains in A1c and TIR at one year alongside a 58% drop in severe hypoglycemia (178). and a Swedish national registry identified Control-IQ as having the greatest A1c reduction among all AID systems (179). In a prospective, non-randomized, real-world comparison with the MiniMed 780G, glycemic outcomes are broadly comparable, though Control-IQ users showed greater improvement in diabetes-related distress (176,180).

TANDEM MOBI

The Tandem Mobi runs the same Control-IQ algorithm but is less than half the size of the t:slim X2. It is FDA-cleared for ages 6 and up. It holds up to 200 units, charges inductively, and is operated entirely through a smartphone app, with only a single on-pump button for quick boluses. A wearable sleeve with 5-inch tubing gives it a more discreet profile than traditional pumps. Since the underlying algorithm is unchanged, the efficacy data from the Control-IQ trials apply, though real-world data specific to the Mobi platform are still accumulating (181).

OMNIPOD 5

The Omnipod 5 is the only tubeless, on-body option with no tubing and no separate pump to carry. The Omnipod 5 is FDA-cleared for T1D for use in individuals ≥2 years of age and for T2D in 2024 (138). The system consists of a wearable pod that contains insulin, and pairs with the Dexcom G6, G7, 15-day G7, and FreeStyle Libre 2 Plus via Bluetooth. Glucose targets in automated mode can be customized from 110 to 150 mg/dL in 10 mg/dL increments. The system is controlled via a smartphone app (both Android and iOS) with an integrated bolus calculator. The pivotal single-arm trial of 241 children and adults with T1D showed meaningful A1c and TIR improvements (182) and a subsequent RCT of 194 adults with suboptimal control demonstrated a 17.5 percentage point TIR improvement over standard pump plus CGM without any severe hypoglycemia or DKA (180). Data in very young children (ages 2 to 5) with two-year follow-up showed sustained glycemic benefits and 97% time in automated mode (148,183). For T2D, a nonrandomized study of 305 adults showed TIR improvement from 45% to 66% with consistent results regardless of insurance status or concurrent GLP-1 or SGLT2 inhibitor use (184). Real-world data from nearly 70,000 users track closely with trial outcomes (185). Retrospective analyses suggest the 780G and Control-IQ have lower TIR than the Omnipod 5 in network meta-analyses (186), though real-world comparative studies show that the differences are modest (187). No RCT data exist in pregnancy and it is not FDA-cleared for use in pregnancy.

In a 13-week RCT of 194 adults with T1D and A1c 7-11%, the Omnipod 5 AID system was compared to sensor-augmented pump therapy. Participants were randomized 2:1 to Omnipod 5 or control. The AID group achieved a significantly greater increase in time in range (17.5% or 4.2 hours/day; $p<0.0001$) and a larger A1c reduction (-1.24% vs. -0.68%; $p<0.0001$). Time spent in hypoglycemia (<70 mg/dL) was lower (1.18% vs. 1.75%; $p=0.005$), and time above 180 mg/dL was substantially reduced (37.6% vs. 54.5%; $p<0.0001$). No episodes of severe hypoglycemia or diabetic ketoacidosis occurred in the AID group. These findings demonstrate that the tubeless Omnipod 5 AID system offers significant glycemic and safety benefits over conventional pump plus CGM therapy in T1D (188).

The SECURE-T2D trial, a prospective, single-arm clinical trial, evaluated the Omnipod 5 AID system in 305 adults with T2D using insulin. The study included a diverse population (57% female, 24% Black, 22% Hispanic) across 21 states, with a mean age of 57 and baseline A1c of 8.2%. After 13 weeks of Omnipod 5 use, A1c dropped by 0.8% (to 7.4%), and TIR improved from 45% to 66% without increasing hypoglycemia risk. These improvements were consistent across all subgroups, regardless of age, race, insulin regimen, or use of adjunct therapies. The study confirmed the safety and effectiveness of Omnipod 5 in T2D and supported its FDA clearance (189).

ILET BIONIC PANCREAS

The iLet Bionic pancreas was approved by the FDA in 2023 for individuals ≥6 years with T1D and is unique in that it requires minimal user input. This insulin pump is initiated using the patient's body weight and requires qualitative meal announcements ("usual for me," "more," or "less") but not formal carb counting and thus represents an incremental step toward a fully closed loop insulin pump. Users cannot modify or override insulin boluses. This system is compatible with the Dexcom G6, G7 and 15-day G7. No warm-up period is required. A 13-week multi-center randomized study of 219 participants with T1D demonstrated a greater reduction in A1c with the iLet bionic pancreas (-0.5%, 95% CI -0.6 to -0.3; p<0.001) but no difference in hypoglycemia compared to standard care (84). In the pivotal NEJM trial of 326 participants ages 6 to 79, the iLet bionic pancreas reduced A1c from 7.9% to 7.3% and improved TIR by 11 percentage points versus standard care, with noninferior hypoglycemia, no DKA, and greatest benefit in those with higher baseline A1c or no prior HCL experience (84). An extension study in participants crossing over from standard care reproduced the same magnitude of benefit (190). A prespecified subgroup analysis found slightly greater glycemic improvements in minority populations (A1c −0.53% vs −0.45%), suggesting potential to reduce diabetes-related health disparities (85). A prespecified psychosocial outcomes analysis also showed significant reductions in fear of hypoglycemia and diabetes distress, with a significant improvement in perceived well-being (81).

SEQUEL TWIIST

The Sequel Med Tech Twiist is a next-generation AID system that received FDA clearance in 2024 for use in individuals ages 6 years and older with T1D (191). It is a tubed pump controlled via the twiist smartphone app on iPhone or Apple Watch, compatible with the FreeStyle Libre 3 Plus and Eversense 365 CGM systems, with remote data sharing available through the twiist insiight app and reporting through Tidepool. The system uses a model predictive control (MPC) algorithm that adapts to individual insulin requirements. It offers a 200-unit insulin reservoir. The algorithm adjusts basal rates every 5 minutes based on a 6-hour predicted glucose, with a correction range programmable between 87 and 180 mg/dL in 30-minute increments throughout the day. Active insulin time is fixed at 6 hours, and no automated correction boluses are delivered. A premeal preset can lower the correction target up to 1 hour before meals, and a workout preset raises the target before exercise to reduce hypoglycemia risk. FDA clearance was based on a single-arm pivotal safety study demonstrating improvements in TIR and A1c with low rates of severe hypoglycemia and DKA.

Table 7.

Comparison of Hybrid Closed Loop Systems

	Medtronic 670G/770G	**Medtronic 780G & Flex**	**Tandem T:Slim X2 & Tandem Mobi with Control-IQ**	**Omnipod 5**	**iLet Bionic Pancreas**	**Sequel Twiist**
Insulin delivery	Tubing	Tubing	Tubing	Tubeless (pod)	Tubing	Tubing
CGM	Guardian 3	Guardian 4, Simplera Sync, Instinct	Dexcom G6, G7, FreeStyle Libre 3 Plus	Dexcom G6, G7, FreeStyle Libre 2 Plus	Dexcom G6	FreeStyle Libre 3 Plus, Eversense 365
Reservoir capacity (unit)	300	300	300 (X2), 200 (Mobi)	200	180	200
Calibration needed	Yes	No	No	No	No	No
Supplies*	DME company	DME company	DME company	Pharmacy	DME or pharmacy	Pharmacy
Control via smartphone	View data	View data (780G)	Bolus only (X2) Full control	Full control	View data	Full control (iPhone)

	Medtronic 670G/770G	**Medtronic 780G & Flex**	**Tandem T:Slim X2 & Tandem Mobi with Control-IQ**	**Omnipod 5**	**iLet Bionic Pancreas**	**Sequel Twiist**
		Full control (Flex)	(Mobi)			
Algorithm initiation	48-hour in manual mode to estimate TDI	48-hour in manual mode to estimate TDI	Weight and TDI entry with maximum delivery of 50% TDI over 2 hr	TDI estimated from programmed basal rates	Weight entry	Basal rates, carb ratios, and ISF
Bolus automation	No	Yes, every 5 minutes	Up to 1 auto-correction bolus/hr if glucose predicted >180 mg/dL	No	Yes	Yes (user defined)
Other inputs	CIR, AIT Unable to override bolus dose	CIR, AIT (2 hour recommended)	CIR, ISF, AIT (fixed at 5 hour)	CIR, ISF, AIT Bolus calculator uses CGM rate of change	“Usual for Me”, “More”, or “Less” customized by meal	CIR, ISF, basal rates, max basal rate; AIT fixed at 6 hours; correction range in 30-minute increments
Extended bolus	No	No	Yes; up to 2 hr	No	No	Yes; via meal types and retroactive carb entry edits
Algorithm adjustment	Every 6 days	Every 6 days	TDI used to scale basal changes	Every 3 days with pod change	Continuously or based on change in entered weight	Continuously every 5 minutes
Target	120 mg/dL	100, 110, 120 mg/dL	112.5-160 mg/dL	110, 120, 130, 140, 150 mg/dL, customizable by time of day	100, 110, 120, 130 mg/dL, customizable by time of day	87–180 mg/dL, by time of day in 30-minute increments; premeal preset lowers target up to 1-hour premeal
Exercise	Temp target 150 mg/dL for 2-12 hour	Temp target 150 mg/dL for 2-12 hour	Exercise target 140-160 mg/dL (cannot program duration) Sleep mode: target 112.5-120 mg/dL without	Target 150 mg/dL, “less aggressive”, for 1-24 hr	None	Workout preset raises target (programmable 87–250 mg/dL);

	Medtronic 670G/770G	Medtronic 780G & Flex	Tandem T:Slim X2 & Tandem Mobi with Control-IQ	Omnipod 5	iLet Bionic Pancreas	Sequel Twiist
			auto-correction bolus, programmable			
Safety mode**	Yes, forced exits from HCL	Yes, exits to Safe Basal up to 4 hours (5.7 vs. 1.7 x/week with 670G)	Not applicable (defaults to basal rate settings)	Yes, forced exits "rare"	Yes, BG-run mode uses manually entered BG up to 72-hour, forces switch to alternate therapy.	Reverts to programmed basal rates in 30 minutes if CGM connection lost > 15 minutes;

PID=proportional integral derivative (system with continual change in response to error between actual and target values). MPC=model predictive control (dynamic reference model serves as a basis. TDI=total daily insulin dose, CIR=carbohydrate to insulin ratio, AIT=active insulin time, ISF=insulin sensitivity factor. *Varies by payer

** Safety mode provides a mechanism to ensure insulin delivery in case of loss of sensor input or threshold for insulin delivery guardrail is reached.

Closed Loop Systems

Additional steps toward closed loop control (CLC) insulin delivery require algorithmic insulin adjustments, which arguably present additional safety concerns (192). Overnight CLC insulin delivery is relatively straightforward, whereas post-meal control and exercise effects remain the most challenging of events to manage. Until recently, most randomized studies have been small and reported only short-term outcomes, often in controlled settings. However, recent advances have led to FDA clearance of systems that move closer to fully automated insulin delivery by eliminating the need for carbohydrate counting (iLet pump), and clinical trials have now evaluated fully closed-loop systems without meal announcement. At this time, there are insufficient data demonstrating the superiority of one system or algorithm compared to others. The three most common algorithms for HCL and CLC systems are:

Model Predictive Control (MPC): predicts future glucose levels and adjusts insulin delivery in response.

Proportional Integral Derivative (PID): calculates the deviation of glucose from target to determine insulin delivery.

Fuzzy Logic (FL): mimics insulin dosing based upon clinical expertise.

DUAL HORMONE SYSTEMS

Systems have utilized single hormone (rapid acting insulin only) or dual hormone (both fast-acting insulin analog and glucagon to imitate normal physiology) as directed by a computer algorithm (Figure 4) (193).

A meta-analysis of 40 randomized studies (35 studies using insulin alone and 9 dual hormone studies) including 1027 participants with T1D demonstrated a significant increase in % TIR (70-180 mg/dL, weighted mean difference 9.6%, 95% CI 7.5-12%), as well as less time in hypoglycemia or hyperglycemia, regardless of type of system (194). Another meta-analysis of 24 studies and 585 participants (7 studies using dual-hormone therapy and 20 studies of insulin only) reported greater improvement in TIR with artificial pancreas systems (12.6%, 95% CI 9.0-16.2, p<0.0001), and greater improvement with dual hormone compared to single hormone systems (195). A meta-analysis of studies with at least 8 weeks duration confirmed these findings (196). A systematic review and meta-analysis of 25 studies in 504 children demonstrated superior %TIR with AID systems vs. open-loop/standard pump therapy with the greatest efficacy observed in bihormonal AID systems, though this conclusion is limited by small subgroup size (N=51) (197). The most comprehensive head-to-head bihormonal analysis by Zeng et al including 17 RCTs found no significant difference in TIR between dual and single hormone systems, though dual hormone modestly reduced hypoglycemia by approximately 17 minutes per day (82). This benefit came at the cost of an 18% absolute increase in gastrointestinal side effects, primarily nausea. Against conventional pump therapy, dual hormone systems demonstrated substantial superiority with nearly 4 additional hours per day in target range. The most consistent advantage for dual hormone systems appeared to be during and after exercise, where it meaningfully reduced hypoglycemia risk, while single hormone systems appear sufficient for overnight glycemic control. At this time, sufficient data does not exist to establish the superiority of any one system over another.

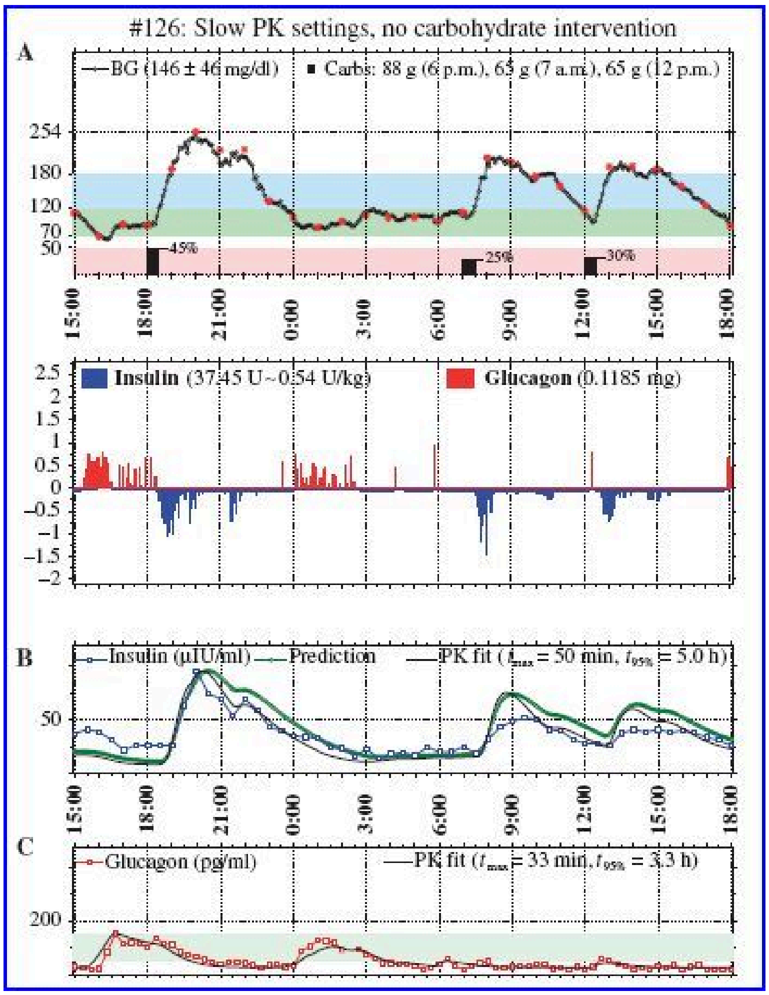


Figure 4.

Dual hormone Closed Loop Control system.

MINIMALLY INVASIVE AND NONINVASIVE GLUCOSE MONITORS

Continuous hypoglycemia detection systems using current sensing technology must be either implanted (fully or partially, either subcutaneously or into a blood vessel). Implantation is more secure but may be associated with biocompatibility problems or local irritation. Less invasive methods may be categorized as minimally invasive or noninvasive. Minimally invasive techniques extract fluid (tears or interstitial fluid) while noninvasive technologies do not. Minimally invasive methods include electrical, nanotechnology, and optical approaches while noninvasive techniques rely on some form of radiation without the need to access bodily fluids. Noninvasive methods frequently incorporate electric, thermal, optical, or nanotechnology methods for detection. Many noninvasive devices under development are aimed for non-continuous monitoring as they often require controlled surroundings including factors such as light, motion and temperature.

> Optical approaches utilize reflective, absorptive, or refractive properties of infrared and optical bands of the light spectrum to detect glucose. Pure optical methods under development utilize Raman and Near infra-red spectroscopy.
>
> Thermal methods detect glucose via the far-infrared band of the spectrum and provide noninvasive approaches for glucose monitoring.
>
> Electric methods use electromagnetic radiation, currents, or ultrasound approaches to detect dielectric properties of glucose. Reverse iontophoresis has been employed with early minimally invasive approaches while bioimpedance spectroscopy has been used in recent noninvasive approaches.
>
> Nanotechnologies aim to miniaturize existing technologies, including fluorescence and surface plasmon resonance (SPR) approaches (198).
>
> Photoplethysmography (PPG) combined with near-infrared technology and machine learning algorithms represents a new approach for blood glucose prediction, with feature extraction from PPG signals enabling glucose estimation without direct fluid sampling (83).

Few devices (other than interstitial CGMs discussed above) have demonstrated high levels of accuracy recommended by expert groups, though a few have been approved by CE or FDA (199,200).

MOBILE TECHNOLOGY AND DECISION SUPPORT

The isolated use of glucose monitoring technologies without a specific plan to address the data provides minimal benefit, particularly among patients with T2D or those not using insulin (201). In order for glucose monitoring to be most effective, data must be easily obtained, organized to identify patterns, and communicated with feedback at the point of care. The widespread use of mobile devices provides opportunities for data collection, analysis, and communication with health care providers, and remote models of care (202). It has also become paramount that devices interface with electronic medical records (203).

Hurdles to wider implementation include the lack of usability, safety, efficacy (including long-term adherence), and cost-effectiveness studies (204). The rapidly changing technology often renders the technology obsolete by the time vigorous clinical trials are conducted and published. The fee for service model is a major barrier, though reimbursement is possible for CGM set-up and professional CGM interpretation (CPT 95251) and via remote physiologic monitoring codes. Finally, cyber security is a big concern for all medical devices, especially for devices that are controlled by a smartphone (205).

Device Downloading, Connectivity, and Interoperability

Manual recording of glucose data is fraught with inaccuracies (206). Most monitors can be downloaded, via a tethering cable or wireless connection, either by the patient or healthcare provider. Each glucose monitoring device generally works with its own proprietary management software. Programs such as Tidepool and Glooko/Diasend can download and organize data from multiple devices, standardizing reports to facilitate efficient review and support population health and telehealth strategies (207). Retrospective weekly data review is associated with improved TIR (208,209) as well as patient reported outcomes including confidence in avoiding hypoglycemia, overall well-being and diabetes distress (210).

Direct connectivity of CGM or blood glucose levels to cell phones improves data integrity and simplifies integration with insulin use, carbohydrate intake, and activity data to facilitate dose adjustments in real time or retrospectively, while also supporting communication with providers. A few meters with direct cellular capability are available. Devices with Bluetooth or cellular connectivity can be paired with apps that support data collection, analysis, and education at the point of care. A regulatory pathway has been developed for alternate controller enabled (ACE) infusion pumps, which can be operated with interchangeable components including CGMs (211).

Diabetes Apps

A variety of stand-alone smartphone applications support glucose monitoring, ranging from simple tracking tools to more sophisticated platforms incorporating machine learning approaches that adapt to the user's previous experiences. Most provide information or track manually entered data, with some supporting insulin or carbohydrate documentation, medication adherence, and physical activity. Apps have shown limited magnitude and sustainability of effect due to user fatigue, reliance on continuous manual data entry, and lack of integration with the health care team. Privacy is an ongoing concern, as federal regulations do not currently prevent app developers from disclosing data to third parties, and expert groups have developed guidance statements to improve standardization (212–214). The ADA recommends prioritizing apps integrated within FDA-approved devices such as CGM or AID systems, or prescription digital therapeutics (PDTs) specifically approved by the FDA for dose titration (1).

Efficacy

Multiple systematic reviews and meta-analyses demonstrate modest A1c reductions of approximately 0.5% with mobile diabetes applications in T2D, particularly among younger patients and those using apps with healthcare provider feedback or wireless data entry (215–219). A 2025 meta-analysis of 41 RCTs confirmed a mean A1c reduction of 0.49% compared to standard care (86), and a 2024 meta-analysis of 57 RCTs further found that apps targeting self-monitoring behavior and medication adherence were associated with the greatest metabolic benefit (220).

Overall efficacy remains variable given heterogeneity in app features and the tendency for apps to be studied as part of multi-component interventions, making it difficult to isolate their independent contribution. Researchers have identified standard evidence-based features that should be included in diabetes apps, such as education, glucose monitoring, and reminders (221,222), though most apps do not address self-management comprehensively, and common usability problems include multi-step tasks and poor navigation (217,223–225). Pattern management software improves provider efficiency in identifying therapeutic adjustments and may provide dosing advice directly to the patient or provider (226,227).

Insulin Dosing Calculators

Bolus calculators are known to substantially improve dosing accuracy and glycemic control in outpatients with T1D (228–230), and may be particularly beneficial for patients with low numeracy. Most incorporate current glucose level, active insulin time, and carbohydrate intake, though few account for activity or illness. Applications that improve carbohydrate counting accuracy are of particular value given that carbohydrate estimation is a major source of dosing error regardless of educational level (231). Connected pens provide insight into missed or altered doses and, when integrated with CGM data, facilitate evaluation of bolus timing. Numerous apps are now available for bolus calculation and basal insulin titration, though only a few have been formally evaluated by regulatory agencies, many require manual data entry, and evidence for basal titration apps is limited (232). Notably, the T1D1 app received FDA clearance in 2025 as the first over-the-counter insulin dosing calculator, following a successful human factors study (233). Other apps operate in conjunction with connected meters and insulin pens subject to regulatory oversight, with functionality ranging from insulin tracking and on-board calculations to bolus dose calculators and adaptive titration (234,235).

Integration within the Electronic Health Record (EHR)

Patient-generated data remains difficult to integrate within the EHR in a clinically meaningful way, resulting in data presented without context and disrupted workflows. CGM data can be integrated via manufacturer platforms or third-party aggregators such as Glooko, Tidepool, Rimidi, or Welldoc, many of which also offer population health tools that flag patients based on CGM data (236,237). Standardization of integration including account linkage, data security, and data granularity remains a significant challenge (238). The iCoDE consensus project established standards whereby a provider order triggers patient consent via electronic message, after which a standardized CGM report is uploaded to the HER (239). Importantly, none of these tools replace frequent patient contact and feedback (240).

BIOMARKERS OF GLYCEMIC CONTROL

Hemoglobin A1c (A1c)

A1c is the best biomarker indicator of glycemic control over the past 2-3 months due to strong data predicting complications (2,10). In addition, the ADA has recommended its use for the diagnosis of diabetes (241).

Hemoglobin A1c refers to the nonenzymatic addition of glucose to the N-terminal valine of the hemoglobin beta chain. Assays are based upon charge and structural differences between hemoglobin molecules (242,243). Therefore, variants in hemoglobin molecules may lead to analytic interferences. It should be noted that some homozygous hemoglobin variants (HbC or HbD, or sickle cell disease) also alter erythrocyte life span and therefore, even if the assay does not show analytic interference, other methods of monitoring glycemia should be utilized, as A1c will be falsely low. Individual assay interferences are available at the National Glycohemoglobin Standardization Program website: www.ngsp.org (244). Several commercial home monitoring kits are also available (245). The two reference methods used to standardize A1c levels are 1) HPLC and electrospray ionization mass spectrometry or 2) a two-dimensional approach using HPLC and capillary electrophoresis with UV-detection (246). A brief summary of assay methods is described below.

> HPLC methods utilize the fact that glycated hemoglobin has a lower isoelectric point and migrates faster than other hemoglobin components. As such it has variable interference with hemoglobinopathies that alter the charge of the molecule (such as HbF and carbamylated Hb), but these may be revealed through individual inspection of the chromatograms.
>
> Boronate affinity methods are based upon glucose binding to m-aminophenylboronic acid and measures glycation on the N-terminal valine on the beta chain but also glycation at other sites. There is minimal interference from hemoglobin variants since the assay does not rely on the charge of the hemoglobin molecule for quantification, but this assay is not widely available. Moreover, the method is still impacted by conditions that shorten red blood cell survival.
>
> Immunoassays make use of antibody binding to glucose and N-terminal amino acids on the beta chain and therefore may be affected by hemoglobinopathies with structural changes at these sites, including HbF but not HbE, HbD, or carbamylated Hb. Some newer assays have attempted to correct for these interferences.
>
> Enzymatic methods lyse whole blood, releasing glycated N-terminal valines which are detected using a chromogenic reaction and are not affected by hemoglobin variants.

An Organization with links to governmental regulatory agencies, the National Glycohemoglobin Standardization Program (NGSP) (<http://www.ngsp.org/news.asp >), evaluates every laboratory and home test for A1c, sets accuracy standards, and certifies which methods meet their standards (247).

A1c is an analyte found within red blood cells, comprised of glycated Hemoglobin. The glycation gap (formerly known as the glycosylation gap) (GG), based on fructosamine measurement, and the Hemoglobin Glycation Index

(HGI), based on mean blood glucose, are two indices of between-individual differences in glycated hemoglobin adjusted for glycemia. GG is the difference between the measured A1c test and the A1c test result predicted from serum fructosamine testing based on a population regression equation of A1c on fructosamine (248). HGI is the difference between the measured A1c test and A1c results predicted from the mean blood glucose level (calculated from self-monitored blood glucose tests or CGM) based on a population regression equation of A1c tests on mean blood glucose levels (249). These two indices are consistent within an individual over time and reflect an inherent tendency for an individual's proteins to glycate (250,251). Patients with high GG and HGI indices might have falsely high A1c test results and might also be at increased risk of basement membrane glycosylation and development of microvascular complications. Whether between-individual biological variation in Hemoglobin A1c is an independent risk factor, distinct from that attributable to mean blood glucose or fructosamine levels, for diabetic microvascular complications is controversial (252).

Because the A1c test is supposed to reflect the mean level of glycemia, attempts have been made to correlate this widely accepted measure with empirically measured mean blood glucose levels. In 2008, the A1c-Derived Average Glucose (ADAG) study compared A1c and CGM derived mean glucose and 7-point glucose profiles among 507 patients with T1D and T2D and without diabetes from 10 international centers to derive an estimated average glucose (eAG) from A1c levels using the following equation: eAG(mg/dL)=(28.7* A1c)-46.7 (Table 8).

Table 8.

A1C and Estimated Average Glucose

A1C (%)	eAG (mg/dL)	eAG (mmol/l)
5	97	5.4
6	126	7.0
7	154	8.6
8	183	10.2
9	212	11.8
10	240	13.4
11	269	14.9
12	298	16.5

Several lines of evidence support this disconnect from a tight correlation between mean glycemia and A1c levels. First, improvements in mean glycemia may not necessarily be reflected by improvements in A1c in intensively treated patients (253). A1c does not reflect short-term changes in glucose control and therefore can be misleading where there have been recent changes in the clinical condition. In addition, glucose fluctuations, compared to chronic sustained hyperglycemia, have been shown to exhibit a more specific triggering effect on oxidative stress and endothelial function (254,255). Glycemic variability cannot be assessed by a global measure of mean glycemia, such as A1c , but requires multiple individual glucose values, such as what can be obtained from continuous glucose monitoring or from seven-point-per-day (or greater) self-glucose testing. Third, A1c does not permit specific adjustments in therapy, particularly among patients requiring insulin titration. Finally, A1c reliability may be affected by several conditions that alter red blood cell lifespan and its use in these circumstances can be misleading. A common example is glucose-6-phosphate dehydrogenase, which is very common in certain groups (African Americans, South Asians) and particularly effects males. A comparison of the features and limitations in glucose markers is presented in Table 9 (256–258).

Ethnic differences in A1c have also been reported (259). For example, data from the T1D Exchange demonstrates a 0.4% higher A1c at a given mean glucose among black patients compared to white patients with T1D, but no effect of race on glycated albumin or fructosamine (260). A 2024 substudy of the GRADE trial involving 1454 participants showed that the relationship between average glucose (measured by CGM) and A1C differs significantly across racial groups, with A1c levels 0.2-0.6 percentage points higher in non-Hispanic Black than in non-Hispanic White patients for average glucose levels from 100 to 250 mg/dL (261). The 2026 ADA Standards of Care note that race and ethnicity should not be considerations for how A1c is used clinically, as genetic determinants that may modify the association between A1c and glucose levels are likely present in only a small fraction of individuals across all racial groups (10).

However, NHANES data do not demonstrate an effect of ethnicity on the association between A1c and retinopathy (262). Data from the ARIC study demonstrated that A1C, fructosamine, glycated albumin, and 1,5-AG were consistent with residual hyperglycemia among blacks compared to whites, and the prognostic value for incident cardiovascular disease, end stage renal disease and retinopathy were similar by race (263). It should be noted that the range of available A1c was relatively narrow in NHANES and ARIC, and further data across an expansive range is needed. In relation to CGMs, utility of A1c is further enhanced when used as a complement to glycemic data measured by CGM (8). Other biomarkers are becoming more widely used; however, A1c remains the most common and well validated biomarker. Other measures of average glycemia such as fructosamine and 1,5-anhydroglucitol are available, but their translation into average glucose levels and prognostic significance are not as clear as for A1c (1).

Table 9.

Comparison of Markers of Glycemic Control

	Biomarker mechanism	**Interval of time reflecting glucose control**	**Cautions/Interferences**
A1c	Hemoglobin glycation	3 months	Hemoglobinopathy/variants (↓/↑*) Decrease in RBC survival (hemolysis, splenomegaly, pregnancy, drugs) (↓) Increase in RBC survival (Erythropoietin, iron replacement) (↑) Transfusion (↓)
Fructosamine/glycated albumin	Protein glycation	2 weeks	Conditions resulting in hypoproteinemia (severe cirrhosis, nephrotic syndrome, enteropathy) (↓) High dose Vitamin C, severe hyperbilirubinemia/uremia/ hypertriglyceridemia (↑)
1,5-AG	Renal clearance	1 week	Chronic kidney disease (stage 4, 5) (↓) Glucosuria (pregnancy, renal tubular disorders SGLT2 inhibitors) (↓) Advanced cirrhosis (↓) High soy diet (↑)

* Assay-dependent.

Glucose Management Indicator (GMI)

GMI is a mathematical derivation from CGM mean glucose, expressed in the same units as A1c (% or mmol/mol), that is now included on all standard AGP reports (11). GMI was developed to simplify interpretation of CGM data and allow for comparison with A1C. However, GMI and A1c values differ in up to 81% of individuals by more than ±0.1% and by more than ±0.3% in 51% of cases (264). In a prospective study of 144 adults with T2D, the correlation between GMI and A1c was only moderate (r = 0.68-0.71), with 36-43% of participants having an absolute difference ≥0.5 percentage points and 9-18% having differences >1 percentage point (265). Overall GMI tends to overestimate A1c at lower levels and underestimate A1c at higher levels. Because of concerns, some experts have called for a re-evaluation of the GMI calculation (266,267). In a recent analysis, an updated GMI was a similar or slightly better predictor of incident retinopathy compared to A1c (268).

Discordance between GMI and A1c can arise from multiple factors including CGM accuracy, sensor calibration, red blood cell turnover, and physiological conditions affecting hemoglobin glycation (269), and a checklist was developed in 2025 to help identify and eliminate sources of discordance (269). The Healthcare Effectiveness Data and Information Set (HEDIS) now permits reporting either CGM-derived GMI or A1c as the metric for assessing glycemic management quality at the health system/population level (11,270).

Time in Range (TIR)

TIR has emerged as an important CGM-derived metric that complements A1c for assessing glycemic control. A 10- to 14-day CGM assessment with ≥70% sensor wear provides reliable TIR data for clinical management (10), with data suggesting a strong correlation between TIR and A1c, with a goal of >70% TIR aligning with an A1c of approximately 7% (11). TIR has been linked to the risk of developing microvascular and macrovascular complications in observational studies; however, prospective data is needed to confirm these findings (91,271). A 2025 narrative review of 45 randomized trials concluded that TIR provides valuable information and should be used as an important endpoint in T2D clinical trials in conjunction with A1c (272).

Fructosamine

A short to medium-term marker (reflecting the average glucose control over the past few weeks) may be useful for determining glycemia over a period of days to weeks since A1c does not reflect recent changes in glucose control. Alternate markers may also be useful in patients with discrepant A1c and self-monitored blood glucose readings as well as patients with other hematologic conditions known to affect A1c. Fructosamine is a term that refers to a family of glycated serum proteins, and this family is comprised primarily of albumin and to a lesser extent, globulinamine. RCTs have reported inconsistent effects of frequent monitoring on A1c lowering, possibly due to differences in execution of therapeutic interventions (273,274). Serial monitoring of short-term markers may also facilitate timely elective surgery in patients whose procedure is delayed due to an elevated A1c. In a recent study, fructosamine was a better predictor of post-operative complications in patients undergoing primary total joint arthroplasty (275).

The 2026 ADA Standards of Care recommend that fructosamine or CGM can be used for glycemic monitoring when an alternative to A1c is required (10). A study of 552 adults with T2D revealed that glycated albumin and fructosamine had similar associations with CGM-defined metrics of hyperglycemia compared with A1c, with c-statistics ranging from 0.85 to 0.94 for detecting TIR and TAR (276).

Glycated Albumin

The largest constituent of fructosamine is glycated albumin. Several investigators and companies are developing portable assays for glycated albumin to assess overall control during periods of rapidly changing glucose levels. In these situations, an A1c test may change too slowly to capture a sudden increase or decrease in mean glycemia. However, there is no RCT showing that the measurement of glycated albumin improves outcomes.

A study in 2023 found that glycated albumin is the most accurate predictor of TIR over 8 weeks compared with other glycemic markers, which could assist in clinical evaluation and medication changes in a more acute setting (277). A comparative study demonstrated that glycated albumin correlates significantly better with both short-term mean blood glucose and A1c compared to fructosamine (278). In the ARIC study, fructosamine, glycated albumin, and 1,5-AG were associated with incident diabetes, even after adjustment for baseline A1c and fasting glucose. In the ARIC study, both fructosamine and glycated albumin predicted incident retinopathy and nephropathy, even after adjusting for A1c (279). However, in adults with severe chronic kidney disease, none of the markers, including A1c , fructosamine, or glycated albumin were very highly correlated with fasting glucose, and there did not appear to be an advantage of one marker over another (280). In addition, baseline glycated albumin and fructosamine were associated with cardiovascular outcomes over a 20 year follow-up period after adjusting for other risk factors, but the overall magnitude of associations was similar to A1c (281). In the Diabetes Control and Complications Trial (DCCT), glycated albumin had a similar association with retinopathy and nephropathy as A1c , but the combination of both markers provided even better prediction (282).

Short-term markers are also of interest for use in pregnancy, where glucose levels are changing more quickly than can be reflected by A1c. Unfortunately, glycated albumin does not predict gestational diabetes more effectively than A1c or fasting glucose (283). However, other preliminary data suggests that glycated albumin may be a better predictor of pregnancy complications than A1c (284).

1,5-Anhydroglucitol

The aforementioned biomarkers for measuring glycemic control, (A1c, fructosamine, and glycated albumin) only reflect mean levels of glycemia. These measures can fail to portray hyperglycemic excursions if they are balanced by hypoglycemic excursions. Plasma 1,5-anhydroglucitol (1,5-AG) is a naturally occurring dietary monosaccharide, with a structure similar to that of glucose (Figure 5). This analyte has been proposed as a marker for postprandial hyperglycemia (285). An automated laboratory grade assay named Glycomark is approved in the U.S. for measuring 1,5-AG as a short-term marker for glycemic control. A similar laboratory assay has been used in Japan. During normoglycemia, 1,5-AG is maintained at constant steady-state levels because of a large body pool compared with the amount of intake and because this substance is metabolically inert. Normally, 1,5-AG is filtered and completely reabsorbed by the renal tubules. During acute hyperglycemia when the blood glucose levels exceed 180 mg/dl, which is the renal threshold for spilling glucose into the urine, serum 1,5-AG falls. This fall occurs due to competitive inhibition of renal tubular reabsorption by filtered glucose. The greater the amount of glucose in renal filtrate (due to hyperglycemia), the less 1,5-AG is reabsorbed by the kidneys. The 1,5-AG levels respond sensitively and rapidly to rises in serum glucose and a fall in the serum level of this analyte can indicate transient elevations of serum glucose occurring over as short a period as a few days.

Measurement of 1,5-AG can be useful in assessing the prior 1-2 weeks for: 1) the degree of postprandial hyperglycemia; and 2) the mean short-term level of glycemia. This assay might prove useful in assessing the extent of glycemic variability that is present in an individual with a close-to-normal A1c level, but who is suspected to be alternating between frequent periods of hyperglycemia and hypoglycemia. In such a patient, the 1,5-AG level would be low, which would indicate frequent periods of hyperglycemia, whereas in a patient with little glycemic variability, the 1,5-AG levels would not be particularly depressed because of a lack of frequent hyperglycemic periods. In the ARIC study, 1,5-AG was associated with severe hypoglycemia after adjustment for other variables, an observation which is consistent with the role of 1,5-AG in reflecting glycemic variability, a known risk factor for hypoglycemia (286).

Longitudinal data from the ARIC study showed that 1,5-AG was associated with ESRD over a 19-year follow-up period, but the relationship was no longer significant after adjusting for glucose control with other markers (287). Among participants with diabetes and A1c <7%, each 5 mcg/mL decrease in 1,5-AG was associated with an increase in dementia risk by 16%, and at A1c >7%, there was also a significant association over a median 21-year follow-up period (288). There was also an association of 1,5-AG and cardiovascular outcomes in ARIC, which persisted, though were attenuated after adjusting for A1c (289). Therefore, it is not yet clear whether 1,5-AG, as a measure of glucose excursions, provides incremental value beyond A1c for predicting long-term complications.

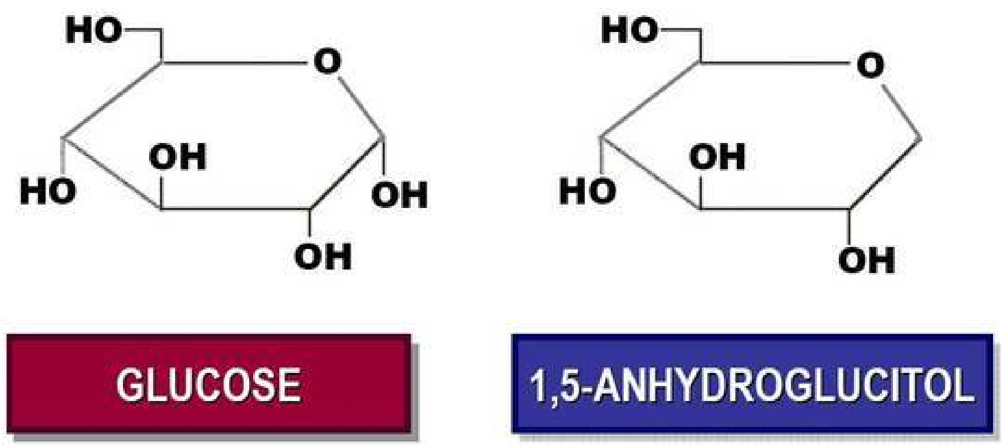


Figure 5.

Structure of glucose (left) and 1,5-anhydroglucitol (right).

KETONE MONITORING

Ketone monitoring complements glucose monitoring by identifying insulin deficiency and evolving ketosis, including DKA. In DKA, blood B-hydroxybutyrate (BHB) is the most relevant ketone, while urine acetoacetate (AcAc) and breath acetone provide alternative signals of ketosis (Figure 6). Ketone testing is particularly important in T1D during illness, persistent hyperglycemia, vomiting/dehydration, or suspected pump/infusion failure, and in people with diabetes taking SGLT2 inhibitors, where ketones can serve as indicators for impending euglycemic DKA. Moreover, ketone monitoring reduces the risk of hospitalization for DKA (290). Despite recommendations, real-world ketone testing remains underused, likely due to a combination of factors including cost, invasiveness, inconvenience, and uncertainty about how to interpret results (290,291).

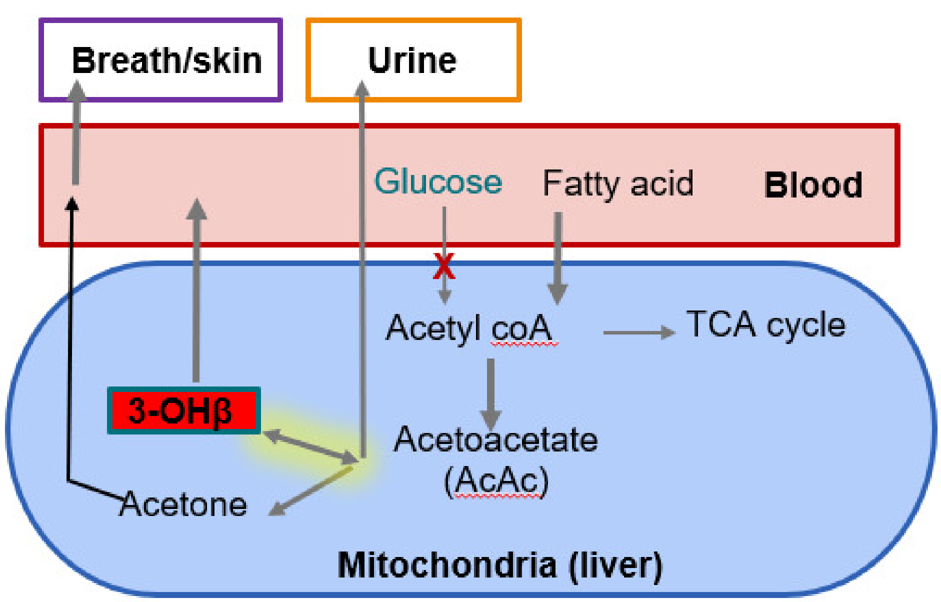


Figure 6.

Formation and distribution of ketones in vivo.

Current Methods

Blood ketone testing (BHB) is the preferred approach for outpatient detection and monitoring of ketosis since it reflects the dominant circulating ketone in DKA and changes rapidly with clinical status. Its main limitations are fingerstick burden and ongoing strip expense (as it is not widely covered by insurance plans), which can reduce adherence. Urine ketone testing (AcAc), while inexpensive and widely available, measures AcAc rather than BHB, may lag behind real-time physiology, is affected by hydration/urine concentration, and can remain persistently elevated during recovery (as BHB converts back to AcAc), making it less reliable for both detection and monitoring of DKA.

Emerging Methods

Continuous ketone monitoring (CKM) via interstitial sensing is a developing approach that could provide early warning analogous to CGM, potentially integrating into existing diabetes technology ecosystems. Data show good agreement with blood BHB levels (292). Thus, the method is under development and may be particularly useful for integration within AID systems, to detect early infusion site failures, and to permit safe use of SGLT-2 inhibitors for cardiorenal benefits in persons with T1D or otherwise ketosis prone individuals. However, there are human factors concerns including patient acceptability with use of multiple sensor sites and potential for alarm fatigue with additional data. Manufacturers are exploring dual sensors measuring both glucose and BHB. In addition, broad adoption will depend on validated accuracy standards, clear clinical workflows for confirmatory testing and escalation and insurance coverage. A recent international consensus report provides objective thresholds for clinical action (293).

Breath acetone is an emerging noninvasive option based on detection of acetone (a volatile product derived from AcAc) in exhaled breath. It may have better clinical utility due to its ease of use and may be more cost-effective since it allows for repeat testing without the need for a new test strip; potentially addressing the adherence gap seen with blood ketone strips. Preliminary data suggest that breath acetone may track blood BHB more strongly during evolving ketosis suggesting its potential use as an early warning tool for DKA detection (294). Otherwise, acetone is limited by lack of strong correlation with BHB. Multiple sensor platforms are under development, including miniaturized chemo-resistive nanosensors. Key challenges include standardized breath sampling (consistent exhalation), environmental sensitivity (humidity/temperature), and establishing clinically actionable thresholds validated against BHB in real-world settings. Currently several breath acetone monitors are available on the market but are primarily marketed for wellness purposes (such as monitoring very low carb “keto” diets) and none are FDA approved for monitoring in persons with diabetes.

CONCLUSIONS

Many new types of technology are increasingly being developed and applied to fight diabetes and its complications. New technologies will improve the lives of people with diabetes by measuring glucose and other biomarkers of glycemic control and linking glucose levels with insulin delivery to optimize glycemic outcomes.

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